

Status, Control, and Risk: How Relative Wealth Perceptions Shape Risk-Taking*

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Abstract

We embed an experiment in a large-scale representative survey to investigate how relative wealth affects risk-taking and how this effect varies as a function of perceived control over life outcomes. Respondents who are induced to perceive their relative wealth as low display more tolerance towards risk in a subsequent incentivized risk-taking task. This effect is not uniform but is mainly driven by individuals who more firmly believe that life outcomes are beyond their control.

Keywords: relative wealth, risk taking, survey experiment, locus of control

JEL classification numbers: C93, D31, D81, D91, E21

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1 Introduction

Relative consumption and relative wealth are essential components of well-being, and economists have long recognized that such positional concerns create an appetite for risk-taking (Friedman and Savage, 1948).¹ Recent theoretical advances have formalized this observation by incorporating status and relative rank directly into utility, with some predicting status-driven risk-taking primarily in the middle of the distribution (Robson, 1996; Becker et al., 2005; Ray and Robson, 2012) and others at the bottom (Kuziemko et al., 2014; Hopkins, 2018). While this disagreement points to differences in where in the distribution relative comparisons matter most, we know little about why individuals may respond differently to the same comparison. Understanding the sources of this heterogeneity is important because individuals differ substantially in how strongly they engage in and respond to relative comparisons (e.g., Clark and Senik, 2010).

We provide empirical evidence on one psychological source of this heterogeneity by examining the role of locus of control, a personality construct that captures individuals' beliefs about the extent to which they can influence events and outcomes in their lives (Rotter, 1966). A large literature shows that personality traits are relevant factors in predicting labor market and behavioral outcomes and shape how individuals respond to economic incentives (see Heckman et al., 2021, for a review). Consistent with this broader evidence, locus of control (LoC) has been linked to labor market outcomes (e.g., Heckman et al., 2006; Caliendo et al., 2015) and risk-related behavior, including household saving, portfolio choice, and gambling (Cobb-Clark et al., 2016; Salamanca et al., 2016; von der Heiden and Egloff, 2021). More directly relevant to our setting, research suggests that people who perceive outcomes as largely beyond their control are particularly responsive to social comparisons, using them to reduce uncertainty and make sense of their environment (Festinger, 1954; Testa and Major, 1990; Aspinwall and Taylor, 1993; Collins, 1996; Fiske, 2011). LoC may therefore moderate the extent to which relative comparisons affect behavior. This heterogeneity can have important economic consequences. For example, Georgarakos et al. (2014) show that perceived peer income contributes to household debt and financial distress among households that consider themselves poorer than their peers. If responsiveness varies systematically with LoC, the same comparison pressure may translate into very different borrowing responses across households.

¹A voluminous literature, dating back at least to Veblen (1899) and Duesenberry (1949), has accumulated evidence that people care not only about absolute outcomes but also their relative position. See for example, Frank (1985); Abel (1990); Gali (1994); Carroll et al. (2000); Ljungqvist and Uhlig (2000); Hopkins and Kornienko (2004); Luttmer (2005); Fliessbach et al. (2007); Card et al. (2012); Cohn et al. (2014); Perez-Truglia (2020); Roth (2026).

Several empirical challenges have hindered credible testing of the heterogeneous relationship between relative wealth and risk-taking. First, meaningfully examining the interplay between relative wealth concerns, risk-taking, and locus of control requires observations on a representative level. Second, one cannot rely on observational data, as varying levels of risk aversion are likely to result in fundamentally different wealth accumulation outcomes to begin with. Third, it is difficult, if not impossible, to create exogenous variation in ranks in the wealth distribution. We address these challenges by designing and implementing a survey module in a high-quality longitudinal study of the German population, the Socio-Economic Panel Innovation Sample (SOEP-IS). With this module, we first measure respondents' perceived ability to control life outcomes and then integrate respondents' objective wealth into an experimental manipulation of relative wealth so that relative comparisons emerge naturally. We use the resulting exogenous variation in perceived relative wealth to estimate its effect on risk-taking and then examine the heterogeneous influence of locus of control beliefs on this relationship.

Our strategy to create the necessary exogenous variation in relative wealth is to induce a shift in respondents' *perceived* relative standing in the wealth distribution. We take advantage of the fact that most people have to infer their relative wealth rank from their limited information on the distribution of income and wealth and systematically misestimate their rank.² In our survey module, respondents had to state their net wealth by placing themselves into one of five wealth categories, and we randomly varied the scale of these categories. Half of the respondents received response categories with wide intervals, i.e., the lowest category ranges up to the 80th percentile of the German net wealth distribution (treatment condition), whereas the other half of respondents received response categories with small intervals, with the lowest category ranging up to the 20th percentile and the highest category starting at the 60th percentile of the German net wealth distribution (control condition). To illustrate the underlying idea, consider a group of respondents with the same objective wealth level, which would place them at the 60th percentile of the wealth distribution. Respondents in the control condition would categorize themselves into the highest category, whereas respondents in the treatment condition would place themselves in the lowest category. Consequently, the randomization of response categories ensures that the objective income and wealth distributions are the same across treatment and control groups, while inducing

²Evidence for the US, for example, suggests that individuals have limited knowledge about the wealth distribution and systematically underestimate wealth inequality (Norton and Ariely, 2011). Relatedly, people tend to underestimate their relative income rank nationally and globally, but adjust these beliefs in response to information about their actual ranks (e.g., Cruces et al., 2013; Karadja et al., 2017; Fehr et al., 2022, 2024).

respondents to feel that their wealth is at the lower end of the distribution and further from the top. We view this exogenous variation in perceived relative standing as a tool to identify how relative wealth affects risk-taking and how this response varies across individuals.

Our analysis proceeds in three steps. First, we show that the treatment affects relative wealth perceptions but not absolute wealth perceptions (“first stage”). In particular, we demonstrate that treated respondents rate their wealth situation as worse relative to other households, think they rank lower in the wealth distribution than non-treated respondents (and are thus less affluent than others), and are further away from the top. Importantly, we find no evidence that the treatment affects absolute wealth perceptions or short-run emotional states, making these alternative channels unlikely explanations for subsequent changes in risk-taking. Taken together, these results indicate that the treatment primarily operates through perceived relative standing in the wealth distribution.

Second, we show that these perceptions matter for risk-taking. We measure risk-taking through a single choice from a set of six lotteries with increasing expected returns and risk (Binswanger, 1980; Eckel and Grossman, 2002). We find that our treatment translates into a sizeable increase in risk-taking behavior. The share of participants choosing the safe instead of a risky option decreases by 23 percent when respondents are induced to perceive their relative wealth as low (in line with theoretical predictions that low relative wealth triggers variation in risk-taking). While this result is reminiscent of findings from a handful of previous studies in small-scale laboratory and lab-in-the-field settings (e.g., Haisley et al., 2008; Kuziemko et al., 2014; Fafchamps et al., 2015; Dijk, 2017; Gamba et al., 2017; Kirchler et al., 2018; Schwerter, 2023), it emerges in a setting where we have shifted the focus from objective to perceived relative standing.³ This distinction is important since it allows us to capture the effects of relative standing in a more natural and representative context, without requiring direct rank assignments or forcing peer comparisons.

To provide a more straightforward interpretation of the shift in risk-taking in terms of standard utility function parameters, we estimate, as is standard in the literature, implied constant relative risk aversion (CRRA) parameters. The estimated CRRA parameter ρ is about 0.58 lower among treated respondents, corresponding to a 20 percent shift relative to the control group mean and implying less risk-sensitive behavior.

Third, we examine more closely the relationship of relative wealth and risk-taking and show

³In these studies, subjects are either assigned a relative rank before making risky decisions, or they make choices involving their lab endowment while a random subset observes others’ earnings or ranks. These designs tightly link the social comparison to the outcomes of the risk task itself, and social comparisons do not emerge organically. An exception is Haisley et al. (2008), who found that low-income individuals ($n = 79$) show a greater tendency to buy lottery tickets when primed to feel relatively poor.

that the response to the treatment depends on respondents' locus of control. LoC is fundamentally a belief about the link between effort and outcomes.⁴ Individuals with a more *external* belief system perceive a weaker link between their own actions and subsequent outcomes, increasing uncertainty about the consequences of their choices. To resolve this uncertainty, they place greater emphasis on relative comparisons because evaluating against others provides valuable information and cues about their situation and position in society (Festinger, 1954; Clark and Senik, 2010; Budría and Ferrer-I-Carbonell, 2019). Since our manipulation induces a perception of low relative standing, this information may carry greater weight for individuals who perceive outcomes as less controllable and are therefore more externally oriented. As a result, they may respond more strongly to treatment. This response may be reinforced if such individuals also view chance-based strategies as a more plausible means of improving their position (e.g., Cobb-Clark, 2015; von der Heiden and Egloff, 2021).

We first present direct evidence on the relationship between locus of control and the tendency to engage in relative comparisons using data from the SOEP-IS. We find that respondents with more external beliefs indeed place greater weight on relative comparisons. Consistent with this pattern, we then show that the effect of the treatment on risk-taking varies systematically with locus of control. In particular, we find a muted response to the treatment from respondents who perceive greater control over outcomes. They also, in general, behave less risk-averse than respondents with external beliefs. In contrast, having external beliefs amplifies the response to perceived low relative wealth: respondents who are more inclined to believe that life outcomes are a matter of luck substantially increase their risk-taking behavior. While the average treatment effect is a 5.4 percentage-point lower likelihood of choosing the safe lottery option, a person with a one-standard-deviation-above-average locus of control score (more external beliefs) responds to the treatment with a 15-percentage-point lower likelihood. Note that in the control group, approximately 23% of all participants choose the safe lottery option.

We are able to address a host of alternative explanations for this result. In particular, we find no evidence that locus of control merely picks up other facets of personal characteristics. Personality traits (Big 5, self-control, and optimism), emotions, and socio-demographic characteristics do not moderate the observed relationship. In addition, we show that the status-risk-locus-of-control nexus is a more general phenomenon. Using data from the World Values Survey, we show that

⁴In this sense it is related to the common luck-effort dichotomy in the inequality and fairness perception literature (e.g., Alesina and Angeletos, 2005; Almås et al., 2020; Fehr and Vollmann, 2025).

perceptions of lower control reinforce the negative relationship between income inequality and willingness to avoid risks.

The study contributes to several strands of the literature. First, we are tying into recent efforts to more rigorously integrate personality traits into economic analysis (Borghans et al., 2008; Almlund et al., 2011; Becker et al., 2012; Heckman et al., 2021; Jagelka, 2024). A central theme in this literature is that individual characteristics, such as the Big 5 and other non-cognitive traits, help explain why people facing the same incentives and constraints make different choices. LoC has received comparatively less attention despite strong associations with key life outcomes, such as labor market success, human capital accumulation, academic performance, savings, or investments (see e.g., Coleman and DeLeire, 2003; Heckman et al., 2006; Barón and Cobb-Clark, 2010; Heckman and Kautz, 2012; Becker et al., 2012; Cobb-Clark, 2015; Caliendo et al., 2015; Cobb-Clark et al., 2016; Salamanca et al., 2016; Pinger et al., 2018; Caliendo et al., 2022). We build on this perspective and study LoC as a source of heterogeneity in behavioral responses to a common economic environment. In particular, we ask whether differences in LoC help explain why the same experience of relative disadvantage gives rise to different risk-taking responses across individuals. Our results highlight that LoC matters: respondents with external beliefs respond more strongly to perceived relative disadvantage and take riskier decisions that may allow them to improve their position relative to others. Taken at face value, these results may have implications for policies that affect how people perceive their economic position, such as income transparency, redistribution, or financial-information policies. While such policies generally cannot be tailored to individuals' control beliefs, their effects may depend on the distribution of these beliefs in the population.

Second, our research advances the existing theoretical literature on status and risk-taking by providing representative large-scale evidence on their competing predictions (see, e.g., Robson, 1996; Becker et al., 2005; Ray and Robson, 2012; Kuziemko et al., 2014; Hopkins, 2018).⁵ These models of status concerns and risk-taking disagree on where in the distribution risk-taking should occur, but commonly focus on objective status and implicitly assume individuals have accurate knowledge of their position in the wealth distribution. Our evidence points to two dimensions that are largely absent from these models. First, our experimental design isolates the causal effect of perceived relative standing while holding objective wealth constant. This reveals that greater

⁵The study also relates to a growing literature on aspirations (see La Ferrara 2019, and Genicot and Ray 2020, for reviews). Aspirations are relative in nature, as comparisons to others significantly inform individual desires and goals. In this sense, aspirations and reference points may encourage risk-taking (Ray and Robson, 2012; Genicot and Ray, 2020; Dohmen et al., 2021; Alaoui and Penta, 2025; Pickard et al., 2024).

risk-taking need not be confined to a particular part of the wealth distribution, as it can arise across a wide range of objective wealth levels when individuals perceive their relative standing to be low. Second, individuals differ systematically in how strongly they respond to such comparisons. In particular, the effect of perceived relative standing on risk-taking varies with beliefs about the extent to which one can control life outcomes. Our results therefore illustrate how personality constructs can affect economic choices through constraints and perceived opportunities that people face (e.g. Borghans et al., 2008).

Third and more broadly, our study contributes to a growing literature on the origins of risk preferences. This literature has, for example, focused on the lasting effects of aperiodic personal experiences such as natural disasters (e.g., Eckel et al., 2009; Cameron and Shah, 2015; Hanaoka et al., 2018), macroeconomic crises (e.g., Malmendier and Nagel, 2011; Cohn et al., 2015) or violence (e.g., Voors et al., 2012; Callen et al., 2014; Jakiela and Ozier, 2019; Brown et al., 2019) on risk preferences. A few other studies investigate short-term fluctuations in risk preferences that are triggered by recurring economic and psychological phenomena such as day-to-day income fluctuations (Akesaka et al., 2023) and emotions (Meier, 2022).⁶ Although we remain agnostic about whether relative position changes underlying risk preferences and instead focus on its effect on observed risk-taking behavior, our study lies between these two strands of research. We complement the literature on violence, crisis, and disaster by explicitly focusing on the impact of a more familiar and natural experience, namely the relative position of individuals in the economic hierarchy of society. To the extent that relative comparisons are frequent, their impact on choices is less transient than, for example, the impact of emotional states.

2 Research Design

We run our study in the Innovation Sample of the German Socio-Economic Panel, the SOEP-IS. The SOEP-IS is a standalone longitudinal study within the SOEP infrastructure that consists of several independent and representative samples. Researchers can propose their own modules in one or more of these samples. Participating households in the SOEP-IS are surveyed yearly, and all household members above age 16 are interviewed in computer-assisted face-to-face interviews by professional interviewers. This has unique advantages over other frequently used survey modes (e.g., phone and online surveys). First, we have very tight control over who is answering the

⁶See Chuang and Schechter (2015) and Schildberg-Hörisch (2018) for a recent review of the literature on variation in (risk) preferences over time.

questions. Second, we can minimize non-responses and clarify misunderstandings on the spot. Third, we can prevent information look-up and communication within households (for more detailed information, see Richter and Schupp, 2015).

2.1 Design of the Survey Module and Measures

We implemented a tailor-made survey module in the 2017 wave of the SOEP-IS. The survey module consists of three parts: questions on locus of control, treatment manipulation, and an incentivized lottery task to measure risk-taking (see the Appendix for the full details of the survey module). Importantly, we explicitly designed the survey module to include these three components, in particular because locus of control was never elicited in our sample in previous waves. Otherwise, we wouldn't have been able to investigate the effect of perceived relative wealth on risk-taking and how locus of control influences this relationship.⁷ Moreover, the sample size was pre-determined by the existing panel structure of the SOEP-IS.

Locus of Control. In the first part of our survey module, we elicit our personality construct of interest, locus of control (LoC), which expresses the extent to which someone believes that life events are under their control (Rotter, 1966). LoC forms early in life and has a significant degree of short- and medium-term stability (Cobb-Clark and Schurer, 2013; Fitzenberger et al., 2021). We implemented the same ten routinely used items in the SOEP (Nolte et al., 1997). For each item, respondents had to agree or disagree with statements such as *“the course of my life depends on me”* and *“which opportunities I have in life is determined by social conditions”*, using a Likert scale ranging from 1 (disagree completely) to 7 (agree completely); see Table A1 for the wording of all items. Because greater perceived control is typically associated with a more optimistic outlook, we will control for optimism (Lerner and Keltner, 2001). We measure optimism with two questions: a general question about optimism regarding the future (on a 4-point scale) taken from the SOEP and a question about the likelihood of experiencing specific events, such as financial gains, career success, or illness, relative to an average person (on a 7-point scale).

Following prior studies, we created a single index for LoC with higher values corresponding to a more external LoC (see e.g. Cobb-Clark and Schurer, 2013; Richter et al., 2013; Specht et al.,

⁷When we started designing the experiment in 2016, pre-registration was not widely adopted outside of development economics (for example, in the AEA RCT Registry, less than 10 percent of all registered RCTs were registered before 07/2016, see registrations in the AEA RCT Registry: <https://doi.org/10.7910/DVN/6TKEJR>). We registered the study ex-post for research transparency as encouraged by the American Economic Association (AEARCTR-0007444).

2013).⁸ Note that LoC is a continuum and that the overwhelming majority of respondents fall between the two extremes (see Figure A1 for the distribution of LoC).

Treatment Manipulation. The second part contains our treatment manipulation. We aim to induce variation in the perceived relative standing in the wealth distribution by asking respondents about their net wealth and varying the response scale of the question. This manipulation builds on research showing that response scales inevitably carry information about the population distribution. In particular, psychological and survey research suggests that respondents use their location on the scale to determine their place in the distribution (e.g., Schwarz et al., 1985; Rockwood et al., 1997; Menon et al., 1997; Bertrand and Mullainathan, 2001).

We explain the concept of net wealth and ask respondents to indicate their net wealth using five predefined wealth categories. We randomly vary the range of available response categories (see Table 1). Half of the respondents receive categories with wide intervals (treatment condition), i.e., *less than €275,000; €275,001 to €468,000; €468,001 to €722,000; €722,001 to €989,000; more than €989,000*. The other half of respondents are assigned to the control condition with much smaller intervals, i.e., *less than €2,500; €2,501 to €11,000; €11,001 to €27,000; €27,001 to €112,000; more than €112,000*. We constructed the intervals based on the most recent data on the German wealth distribution at the time of the survey.⁹ The lowest interval in the treatment condition covers 80 percent of the German net wealth distribution, implying that most respondents should locate themselves at the lower end of the wealth distribution. In contrast, the responses in the control condition should be more concentrated in the middle and top categories.

Risk-Elicitation Task. After the wealth categorization, we measure respondents' willingness to take risks through a single choice from a set of six lotteries (Binswanger, 1980; Eckel and Grossman, 2002). Each lottery from this set has an equal chance of yielding a low or a high payoff. While the first lottery guarantees respondents a payoff of €50, the remaining lotteries gradually decrease the low payoff and increase the high payoff (see Table A2). The lotteries gradually increase in expected value but also in the implicit level of risk (standard deviation). This choice task is easy

⁸The ten LoC items aim to capture four domains: internal and external locus of control (8 items), fairness (1 item), and social involvement (1 item), see Nolte et al. (1997). Table A1 shows the wording and descriptives for each item and outlines how we constructed the index using the seven items with the highest internal consistency (as suggested by Richter et al., 2013, and Specht et al., 2013).

⁹The upper bounds of the intervals correspond to the 20th, 30th, 40th, and 60th percentile of the distribution in the control condition and the 80th, 90th, 95th, and 97th percentile in the treatment condition. The median net wealth in 2014 was 60,400 euros, while the average was 214,500 (see HFCS, Deutsche Bundesbank, 2016, for more information).

to understand and, therefore, well suited for eliciting risk preferences in a general population sample (Dave et al., 2010; Charness et al., 2013), yet still rich enough to obtain detailed utility information.¹⁰ We incentivized the task by randomly selecting one in ten respondents and paying them the outcome of their chosen lottery.

Discussion of the Design. Our design induces an exogenous shift in perceived relative standing in the wealth distribution to identify the causal link from relative wealth to risk-taking.¹¹ This setup ensures that, by design, the actual distribution of income and wealth is identical across treatment and control groups, effectively holding absolute wealth constant. The resulting changes in perceptions and behavior are typically short-lived, which is common in this type of intervention (e.g., Cavallo et al., 2017; Bottan and Perez-Truglia, 2022).¹² This is sufficient for our primary objective, since the theoretical models predict that changes in relative wealth lead to changes in current risk-taking rather than persistent changes in underlying risk preferences. Moreover, relative comparisons arguably arise repeatedly in everyday life, so even temporary shifts in perceived relative standing may affect a range of immediate economic decisions.

Importantly, in our design, respondents receive information about the wealth distribution in the control *and* treatment groups. This has several advantages over an alternative standard design that would elicit prior beliefs about wealth ranks and then inform a random subset of respondents about their true rank to “correct” their prior beliefs. In these designs, the direction of the treatment effect is endogenous to respondents’ priors. First, in contrast to such designs, our manipulation induces a monotonic shift in perceptions within the treatment group, affecting a large portion of the sample uniformly. Second, the treatment variation in perceptions comes from differences in the information about the wealth distribution instead of exogenous changes in prior beliefs. Third, and relatedly, the treatment variation is orthogonal to prior perceptions about relative wealth and other unobserved and observed individual characteristics. This is important for analyzing heterogeneity in LoC as perceived relative wealth may depend on individual characteristics (such as LoC). To

¹⁰A comprehension check after the risk elicitation task revealed that about 89 percent of respondents rated the comprehensibility of the risk-elicitation task as good or very good, and only 2 percent of respondents as insufficient or unintelligible.

¹¹Note that our intervention does not explicitly ask about a relative comparison, but rather a self-categorization on a scale, such that the exact reference group is less relevant. Evidence from the few existing papers suggests that relative comparisons within smaller reference groups are somewhat more consequential for outcomes than larger groups (e.g., Fehr et al., 2022; Hvidberg et al., 2023; Xu et al., 2026).

¹²The pass-through rate of information on beliefs is typically substantially lower than the Bayesian benchmark and fades out within a short time frame. Given that effect sizes of behavior are usually smaller than effect sizes on belief updating, we do not expect that changes in risk-taking are permanent (see Haaland et al., 2023, for a review).

analyze this heterogeneity, we elicited LoC prior to the treatment intervention. This avoids running the risk that imbalances in LoC drive the results and rules out the possibility that the treatment influences responses to the LoC questions.

2.2 Data

Main Sample. Our sample in the SOEP-IS consisted of 1,115 individuals (Sample I3). Because our study requires that respondents participate in the risk-preference elicitation task, and they could refuse to do so, we include only respondents with non-missing observations, resulting in 914 observations. Importantly, missing observations are not more likely to appear in either of the two treatment conditions (17 percent in the control and 20 percent in the treatment condition, t-test, $p=0.20$).¹³ In Section 4.3, we provide further evidence that there is no differential selection into the risk elicitation task and that our results are robust to re-weighting the study sample to match the socio-economic characteristics of the entire sample.

Covariates. We consider a set of observables that have been shown to relate to risk aversion, including age, gender, education, parents' education, marital status, household size, employment status, satisfaction with health and life, region (East/West Germany), income, and wealth (see Dohmen et al., 2011). To measure a respondent's objective wealth, we use household information on assets and housing to estimate household wealth, following Fuchs-Schündeln and Schündeln (2005). As a proxy for wealth, we also consider homeownership, which is the most important component of wealth in Germany below the top 1% of the wealth distribution (Schröder et al., 2020). Finally, we control for baseline risk aversion using a qualitative measure of general risk aversion on a scale from 0 to 10 (very willing to not willing to take risks) before treatment, as well as the Big 5 personality traits and our collected measures of general optimism. We include these covariates to assess whether the heterogeneity associated with locus of control simply reflects other observable individual characteristics.

Balance. In Table A3, we present p-values from a set of linear regressions assessing the balance of observables by treatment status. The results of these regressions (including an F-test for joint significance of all variables) indicate that the treatment and control groups are very similar along

¹³ Respondents could not condition their participation on the lottery stakes because they only received some general explanation of the elicitation task before they had to indicate their participation decision.

these observables. In Table A4, we present the same exercise for LoC and other personality traits. Again, the table indicates that these traits are well-balanced across the treatment and control groups.

Manipulation Check. We rely on three instruments to illustrate through which channel our intervention affects risk-taking: a question in our SOEP-IS module and two manipulation checks in two different study samples. For the two manipulation checks, we recruited two new samples through respondi/Bilendi, an internationally well-known panel provider, and implemented two short surveys (median duration of six minutes) with the same treatment intervention as in our SOEP-IS module (see Section A.1 for more implementation details). In the first survey ($N = 987$, respondi), we measured the impact of the treatment on the perceived rank in the wealth distribution, LoC, self-esteem, and credibility of the wealth data. In the second survey ($N = 900$, Bilendi), we measured the treatment effect on perceived absolute wealth, treatment-induced emotional states, and experimenter demand. The two separate surveys allow us to conduct comprehensive manipulation checks without risking contamination of the outcome by the first stage, or of the first stage by the outcome or the manipulation.

3 First Stage Results

As a first pass to check whether the manipulation worked as intended, we examine the responses to the wealth categorization across both conditions in Table 1. In the control condition, the distribution of responses in the five wealth categories is well dispersed, and a majority of respondents aligned themselves in one of the three top categories starting from €11,000. In strong contrast, about 80 percent of respondents in the treatment condition placed themselves in the lowest category. The categorizations also reflect respondents' objective wealth that we estimated from household information on assets and housing in the SOEP-IS. For example, the average net wealth in the lowest treatment category (ranging to the 80th percentile) is about €127,000, comparable to the average wealth of €126,000 in the four lowest control categories that range up to the 60th percentile. Similarly, the average wealth in the highest control category is close to the level in the four highest treatment categories (€427,000 vs. €370,000).¹⁴ Accordingly, the wider wealth intervals of the treatment condition induced most respondents to locate themselves at the lower end of the wealth distribution.

¹⁴We see a similar pattern if we look at homeownership as a proxy for wealth, which is not surprising given the strong correlation between wealth and homeownership ($\rho = 0.58$).

Next, we examine whether the manipulation induced a feeling of lower relative wealth and being further away from the top. Table 2 displays the results. To start with, we find no evidence that the treatment affects how respondents perceive their absolute wealth. Treated respondents perceive their absolute wealth level similarly to non-treated respondents (column 1), view their wealth as equally adequate (column 2), and include a similar number of risky assets (plus homeownership) in their wealth estimates (column 3).¹⁵ In contrast, we observe that the treatment affects perceptions of relative wealth. First, treated respondents perceive their wealth situation relative to other households as significantly worse than non-treated respondents (column 4). Second, we quantify this assessment by directly asking respondents about their position in the wealth distribution. Column 5 shows that treated respondents rank themselves lower in the wealth distribution than non-treated respondents and thus perceive themselves as less wealthy. Third, we ask respondents in our SOEP-IS sample to estimate the household income above which they are richer than 90 percent of other households.¹⁶ Given the positive correlation between wealth and income, we should also see a qualitative effect on this relative income estimate if the treatment successfully induced a perception of lower relative standing. That is, inferring the 90th income percentile based on one's own income should result in higher estimates in the treatment condition. Column (6) of Table 2 confirms this hypothesis and indicates a strong effect: treated respondents estimate a 36 percent higher income at the 90th percentile of the household income distribution than untreated respondents, indicating a lower relative standing and a larger gap to the top. Together, these findings further illustrate that the intervention affects relative, but not absolute, wealth perceptions.

In addition, we can rule out other channels through which the intervention may influence risk-taking. First, we show that respondents find the presentation of wealth categories plausible in both treatment and control groups. We see virtually no difference in the credibility rating of the presented wealth data (see Table A5, column 1). Second, we test whether the treatment has a bearing on respondents' LoC. We find no evidence for such an effect: treated respondents display a similar level of LoC as respondents in the control group (Table A5, column 2). Note that we collected this post-treatment measure of LoC as part of our manipulation check (see also Section 2.2), whereas the LoC measure for our main analysis was naturally elicited before the treatment

¹⁵Specifically, we asked respondents (i) to describe their household's net wealth (very low to very high on a five-point scale), (ii) to indicate how adequate their household's net wealth is for their needs and plans (not at all adequate to fully adequate on a five-point scale), and (iii) to indicate all assets from a list that they included in their wealth estimate.

¹⁶We placed this question in a subsequent and unrelated survey module and focused on income to obfuscate the link between our treatment and the manipulation check. Douenne et al. (2024) present evidence that people infer their wealth rank from their income rank and vice versa.

intervention. Finally, we address a concern that the treatment induces a temporary change in emotional states, for example, because one learns that one’s wealth is at the lower end of the distribution. We examine the impact of our treatment on positive (self-esteem and happiness) and negative affective states (anger, worry, and sadness).¹⁷ We find no indication that the treatment results in changes in positive self-assessment, happiness, and negative affect (columns 3–5, Table A5). The affect balance indicates that the treatment tends to induce more negative affect. However, the effect is small (0.08 of a standard deviation) and insignificant (column 6, Table A5).

In summary, we take these results as evidence that exposing respondents to wider wealth intervals in the treatment condition generated a strong “first stage” and had the intended effect of significantly shifting respondents’ views about whether their relative standing in the wealth distribution is low or not.

4 Main Results

4.1 Average Treatment Effect

We now analyze how perceived relative standing in the wealth distribution impacts respondents’ willingness to take risks. Figure 1 plots the distribution of lottery choices separately by treatment status. Relative to the control group, fewer treated individuals chose lottery 1, which offered a safe payoff of €50, while the remaining mass was spread across the other five lottery categories. In fact, while 23.0 percent of participants in the control group chose lottery 1 (safe lottery), the corresponding figure in the treatment group was 17.7 percent, a 23 percent decline. Regressing a binary indicator for choosing the safe lottery 1 on a treatment dummy demonstrates that this decline is statistically significant (Table 3, column 1).

This non-parametric result suggests that a perceived low relative standing increases the willingness to take risks: the treatment induces respondents to pick lotteries with higher implicit risk. Column (2) of Table 3 shows that this is more generally true when considering the full distribution of lottery choices and estimating an ordered probit regression. As is also evident from the raw distributions in Figure 1, participants tend to pick higher (riskier) lotteries when their perceived relative standing is low. Interestingly, while lotteries 5 and 6 have the same expected value (€100), lottery 6 has a larger upside potential (€190 vs €200) and is more frequently chosen

¹⁷We asked respondents to think about the question regarding their household’s net wealth and to rate, on a scale from 1 (not at all) to 5 (very strongly), how strongly they were currently feeling the four emotions – happiness, anger, worry, and sadness.

in the treatment group. This reinforces the notion that participants who perceive their relative standing as low are less discouraged by the downside risks of their choices and more inclined to orient their behavior toward maximizing the larger potential payoff.¹⁸

To further characterize the magnitude of the treatment effect, we follow the previous literature (e.g., Binswanger, 1980; Andersen et al., 2008; Brunnermeier and Nagel, 2008; Chiappori and Paiella, 2011; Sahm, 2012; Carvalho et al., 2016a) and use the incentivized lottery choices to infer implied risk-aversion utility parameters. Under the assumption that individuals exhibit constant relative risk aversion (CRRA), we can map each lottery choice into an interval of CRRA parameters. For instance, if the utility of a monetary amount x is given by $u(x) = \frac{x^{1-\rho}}{1-\rho}$, then $\rho = 0.812$ would imply indifference between choosing lottery 3 and lottery 4, whereas $\rho = 1.74$ would imply indifference between choosing lottery 2 and lottery 3, i.e., the choice of lottery 3 would be consistent with $\rho \in [0.812, 1.74]$.¹⁹ We then model ρ_i as an individual-specific latent variable and estimate interval regressions using the link between each lottery choice and the theory-consistent interval for ρ_i .²⁰ The advantage of this approach is that the theory-informed interval regression attaches a cardinal meaning to the underlying latent variable, making the estimate interpretable in terms of standard economic parameters.

Column (3) of Table 3 contains the result. The estimate points to a reduction in the measured CRRA parameter of roughly 0.58. Compared to the control group mean of 2.86, this amounts to a change of about 20 percent. At the same time, we must keep in mind that the average predicted CRRA parameter in both the treatment and control groups is well above zero and thus consistent with risk aversion. While the treatment is consistent with a considerably lower level, it does not, on average, lead to risk-neutral or even risk-loving behavior. This is in line with the relatively large shift from choosing the safe lottery to any of the remaining five lotteries with uncertain outcomes. To put the result into perspective, Carvalho et al. (2016b) estimate that a random assignment to savings accounts for rural Nepalese households leads to a 17 percent increase in risky choices in a similar lottery task. Other studies estimate gender differences in CRRA parameters of 30 percent (e.g., Garbarino et al., 2011). Given these numbers, the average treatment effect presented above

¹⁸See Table A6 for the marginal effects of the ordered probit model.

¹⁹This assumes, as is common in the literature, that people view the financial gains of these incentivized lotteries in isolation and do not integrate their background wealth when making their choice (e.g., Andersen et al., 2008). Otherwise, their choices in small-stakes lotteries would imply implausibly high levels of risk aversion (Rabin, 2000).

²⁰Note that this estimation procedure is tightly linked to the ordered probit model. The ordered probit model assumes that for any lottery choice k , $y_i = k \iff \kappa_{k-1} < \tau \times \text{Treatment}_i + X_i\beta + \epsilon_i \leq \kappa_k$ with $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ and estimates κ_k explicitly. The interval regression model, by contrast, uses the indifference conditions implied by CRRA utility to replace the boundaries κ_k with their corresponding CRRA parameter values. We then estimate only (τ, β) via maximum likelihood.

appears economically meaningful.

There are, however, some caveats when interpreting the results through the lens of CRRA parameters. The underlying lottery consists of six choice options and is consequently coarse. While this simplicity greatly facilitates the measurement of risk-taking behavior in general population samples, it introduces some noise in the elicitation process. The 90% confidence interval of our estimates does, however, contain only negative values (-1.04, -0.13), indicating a shift towards more risk-taking behavior, and the elicited CRRA parameters correlate highly with both previously surveyed qualitative measures of risk-aversion and socio-demographic factors that are typically correlated with higher risk-aversion (e.g., being female or having a lower educational attainment; see Figure A2 and Table A7). Second, we view the estimation of CRRA parameters as a measurement tool for assessing the magnitude of the shift in risk-taking in response to the treatment rather than interpreting the estimated lower CRRA parameters as changes in risk preferences.

Taken together, the results in this section provide evidence that when respondents perceive their relative standing to be low, their lottery choices display substantially increased tolerance for risky behavior. That is, their behavior is consistent with context-dependent risk behavior as, for example, studied theoretically by Kőszegi and Rabin (2007).

4.2 Moderating the Effect: Relative Wealth and Perceptions of Control

This section turns to our second contribution and examines whether and how the response to relative comparisons varies across individuals. In particular, we focus on respondents' beliefs about how life outcomes are determined. Individuals with more external beliefs typically perceive a weaker link between effort and outcomes. That is, they face greater uncertainty about how their actions translate into future outcomes and thus their economic position. One way to resolve or reduce this uncertainty is to rely more strongly on social comparisons, which provide an external benchmark for evaluating one's own position and outcomes. Psychological research suggests that external beliefs are linked to a higher tendency to engage in relative comparisons (Wills, 1981; Wood and Taylor, 1991; Clark and Senik, 2010; Budr  a and Ferrer-I-Carbonell, 2019). We provide direct evidence for this link using data from a different SOEP-IS sample. Figure 2 illustrates that the inclination to compare oneself to others increases with more external control beliefs.²¹ If individuals with more external control beliefs place greater weight on social comparisons, the experimentally

²¹We draw here on the Social Comparison Scale that measures an individual's disposition towards making social comparisons. This scale is only part of a different, independent SOEP-IS sample ($N = 1,388$), and thus we cannot link the scale directly to our respondents (see the notes in Figure 2 for more detail).

induced perception of low relative standing should be more consequential for their choices. We therefore expect that the response to the treatment is stronger among more externally oriented individuals and weaker among those with more internal control beliefs. These heterogeneous effects may be reinforced if externally oriented individuals also view chance-based strategies as a more plausible means of improving their position, which aligns with their weaker perceived link between effort and outcomes (Cobb-Clark, 2015; von der Heiden and Egloff, 2021).

LoC Summary Statistics. The median LoC score in our sample is 3 (average: 3.13, see Figure A1 for the distribution of LoC). The distribution of LoC indicates that the majority of respondents have internal control beliefs, which confirms previous findings from the SOEP (e.g., Nolte et al., 1997). In Table A7 (columns 5–6) and Figure A2, we present correlates of LoC based on control group data. Consistent with existing empirical evidence, we observe an education and income gradient: more education and higher income are associated with lower external control beliefs.

Treatment Effects. Figure 3 displays the distribution of lottery choices by treatment group, separately for those with above and below median LoC. First, at baseline (i.e., focusing on control-group observations only), LoC appears negatively associated with risk-taking behavior. That is, while 19 percent of control-group participants with internal control beliefs chose the safe lottery, 28 percent of those with external control beliefs did. Absent any treatment, we thus see a lower tendency for risky behavior among those with external LoC, confirming the commonly observed pattern in the literature that respondents with an external LoC exhibit higher risk aversion than respondents with an internal LoC.²²

At the same time, Figure 3 shows that while lottery choices are very similar across conditions among those participants with below-median LoC (internal control beliefs), the treatment shifted choices substantially among those with above-median LoC (external control beliefs). As with the sample average, this is again reflected in a lower likelihood of choosing the safe lottery. Columns (1) and (2) of Table 4 display point estimates of linear probability models for the split samples. While the treatment effect is not statistically significant for those with an internal LoC, the probability of choosing the safe lottery drops by 13.6 percentage points in the external LoC sample. Compared

²²Despite group-level differences in *baseline* risk-taking behavior, both groups are overall risk-averse. The average chosen lottery among control group participants is 3.3 for those with below-median LoC and 2.9 for those with above-median LoC. This implies that subgroup-specific ceiling effects, i.e., a disproportionately high fraction of participants choosing the riskiest lottery already in the control group setting, are unlikely to drive the conclusions of our heterogeneity analysis.

with a control group mean of 28 percent, this represents a reduction of approximately half.

In Column (3) of Table 4, we interact the treatment indicator with our standardized and continuous measure of control beliefs, where higher values of LoC are associated with more external beliefs. A one-standard-deviation shift towards more external control beliefs leads to a difference in the corresponding treatment effect that is almost twice the average treatment effect. The same is true when using interval regressions with the implied CRRA curvature parameter as the outcome variable (Columns 4 through 6). For an individual with a LoC score that is 0.5 standard deviations higher, the treatment shifts the CRRA parameter downward by approximately 1. In terms of model calibration, this is equal to the difference between a relative risk aversion parameter $\rho = 2$ and log-utility.²³ Note that the LoC score has a standard deviation of 0.87 and consists of seven items. Half a standard deviation is thus, for instance, consistent with agreeing three points more to the statement “I have little control over the things happening in my life” on a scale from 1 (disagree completely) to 7 (agree completely).

Overall, LoC strongly moderates the effect of perceived relative standing on risk-taking. More externally oriented individuals respond much more strongly to the treatment than those with more internal control beliefs.

4.3 Robustness

Next, we describe a battery of exercises to probe the robustness of our results. These exercises address concerns about internal validity and include alternative specifications of LoC. Importantly, we provide evidence that other personality traits are unlikely to moderate the effect on risk-taking. Finally, we illustrate the broader relevance of the observed heterogeneity in the relationship.

Non-Compliance. Recall that participation in the risk-elicitation task was voluntary. Consequently, a potentially worrisome threat to internal validity is differential selection into the risk-preference elicitation task. We address this issue in several ways. First, we note that the sample is balanced on a set of important observables, including baseline risk aversion (see Table A3). Therefore, it is unlikely that the main effects are driven by differential selection with respect to pre-treatment risk aversion. Second, in Table A10, we provide further evidence that non-participation in the risk-elicitation task is orthogonal to the randomly assigned treatment status. In particular,

²³These quantitatively large changes are confirmed when using an ordered probit model rather than relying on the CRRA classification; see Appendix Table A6 for marginal effects and Table A8 for point estimates.

we show that non-participation in the risk-elicitation task does not differ across conditions (panel a.) and that baseline risk aversion is the same across respondents and non-respondents (panel b.) and treatment and control (panel c.). Third, we show that our results are robust to re-weighting the study sample to match the socio-economic characteristics of the full sample. That is, we use a probit regression to predict the probability of sampling for each respondent based on a set of observable characteristics, and use the inverse of this probability to construct individual weights. Table A11 reveals that this reweighting exercise yields results very similar to our unweighted estimates. Together, there is little reason to worry that the decision to participate in the lottery task biases our estimates.

Functional Form and Alternative LoC Scores. We have shown that treatment effects are substantially more pronounced for respondents with a more external LoC. We test whether this effect is driven by LoC outlier observations along two margins. First, we show that higher-order interactions between the treatment indicator and the LoC score are not significant (see Table A9, columns 1-3), suggesting that the linear interaction model is a plausible approximation and that the effect is widespread across the sample. Second, we show that the results are unaffected when excluding respondents with particularly strong internal and external LoC beliefs (the bottom and top 10 percent of the distribution; see Table A9, column 4).

Following prior studies, we use a single index for LoC in our analysis. However, a concern is that this index is based on unweighted averages (Cobb-Clark and Schurer, 2013; Piatek and Pinger, 2016). We present two alternative measures for LoC to show the robustness of our results; see Table A1 for details. First, we perform a principal component factor analysis to extract separate, weighted scores for internal and external control beliefs (see, e.g., Piatek and Pinger, 2016). We then interact each of the two scores with our treatment indicator. Second, we combine all items from the two separate indices into a weighted single index and interact it with the treatment. In Table A12, we show that the choice of how to represent individual LoC beliefs does not affect our conclusions in a meaningful way.

Other Moderators. We address three sets of potential other moderators that might be correlated with a person's LoC and therefore spuriously picked up by the interaction between our treatment indicator and the LoC measure.

First, a widespread concern is that LoC merely captures the effects of other facets of

personality, such as the Big 5, optimism, or self-control. For instance, Judge et al. (2002) consider the LoC as part of a higher-level construct intermeshed with other traits, such as neuroticism. Similarly, evidence suggests that optimism is related to LoC and risk aversion (Lerner and Keltner, 2001; Dohmen et al., 2018). While our main results already condition on such other personality traits, we take the notion of alternative moderators seriously and fit models that interact our treatment with each component of the Big 5, our optimism proxy, and a self-control score.²⁴ Table A13 shows that none of the alternative traits are stand-ins for LoC. In contrast to LoC, none of the interactions between these other personality traits are statistically distinguishable from zero.

Second, there is some evidence that risk-taking and LoC are tightly linked to emotions (e.g., Loewenstein et al., 2001; Cohn et al., 2015; Meier, 2022). We investigate this issue along two margins. First, we examine whether the treatment has a direct effect on emotional states. As already discussed in Section 3, we find no evidence that the treatment results in changes of positive and negative affect (see Table A5). Second, we consider the possibility that emotions moderate the treatment effect on risk taking. To this end, we draw on the standard emotion module in the SOEP-IS, which was administered before our module, and follow the same approach as in our analysis of personality traits above. In Table A14, we control for anger, fear, and a combined measure of happiness and sadness.²⁵ Again, we basically see that our main result is robust to controlling for emotions. Next, we interact each emotion with our treatment and report the results in Table A15. We find no indication of heterogeneity in our results that are related to emotions. Taken together, this suggests that emotions play little role in our context: neither as an alternative channel for the treatment nor as moderators.

Third, we explore whether the heterogeneous effects of LoC merely reflect heterogeneity in certain socio-economic characteristics. We rerun our estimation using socio-economic characteristics associated with LoC, such as unemployment status, income, and education. Differentiating respondents along these categories also reveals substantial differences in their tolerance of risk. This is, for example, true for gender: women are more risk-averse than men.²⁶ However, we find

²⁴The optimism proxy we use here is based on the following question that we administered before the LoC questions: “If you think about the future: Are you... (1) optimistic; (2) rather optimistic than pessimistic; (3) rather pessimistic than optimistic; (4) pessimistic?”. The conclusions do not change if we instead use a measure for optimism relative to one’s peer group. Note that Dohmen et al. (2018) use a different measure of optimism based on self-reported degrees of optimism and pessimism. The self-control score is based on a 13-item module following Tangney et al. (2004), administered in an independent study of the same sample (Cobb-Clark et al., 2019). We recode each item so that higher values indicate a stronger sense of self-control. Subsequently, we standardized their sum using the sample mean and standard deviation.

²⁵Following the literature on emotions and risk attitudes, we combine happiness and sadness into an index by averaging their difference and adjusting the index to match the other emotions’ ranges (see Meier, 2022).

²⁶The sample gender difference in the implicit CRRA parameter is roughly 33 percent, and the unemployed/employed

no evidence that any of these socio-economic characteristics interact with the treatment, suggesting that LoC affects heterogeneity independent of these characteristics (Table A16).

In the most restrictive and final test, we estimate the interaction between our treatment indicator and LoC in a specification that also includes the interactions between treatment and the full set of covariates included in our main analysis. Although this specification is statistically demanding, the estimate of the interaction effect remains virtually unchanged (0.92 vs. 0.89) and statistically significant (see Table A17).

Statistical Inference. Given the large number of alternative moderators and corresponding hypothesis tests, we also probe the robustness of our approach to inference when studying treatment effect heterogeneity. The interaction between LoC and our treatment remains statistically significant when relying on randomization inference and when controlling for the *false discovery rate* following the two-stage linear step-up procedure proposed by Benjamini et al. (2006). For details, see Section A.3.

Inequality. We provide evidence that our findings extend beyond our context. Relative position in the wealth (and income) distribution is naturally linked to inequality because the distribution affects how much richer others are. At the aggregate level, this notion of relative deprivation is related to the Gini coefficient (Yitzhaki, 1979; Hey and Lambert, 1980). Using data from the World Values Survey, we show that greater income inequality is associated with lower risk aversion and that external control beliefs moderate this relationship (see Section A.2 for more details).

5 Conclusion

We show that perceived relative wealth causally affects risk-taking. Respondents are willing to take more risk when they perceive their relative standing in the wealth distribution as low. This average effect masks substantial heterogeneity. The increase in risk-taking is concentrated among respondents with more external control beliefs, whereas respondents with more internal beliefs show little response to the same relative-wealth manipulation. Thus, the behavioral consequences of perceived relative disadvantage depend systematically on individuals' beliefs about their control over life outcomes.

contrast is 29 percent. Both magnitudes lie well above the difference observed for LoC.

These results highlight the importance of identifying when *and for whom* relative wealth matters for risk-taking. First, they help to refine predictions about risk-taking in response to relative comparisons. Existing theories primarily differ in where in the wealth distribution status concerns generate risk-taking. Our findings point to an additional source of heterogeneity as status concerns need not translate uniformly into greater risk-taking. The behavioral consequences of perceived low relative standing depend in part on how individuals perceive the link between their actions and outcomes. Second, the results also have implications for policies such as income transparency, redistribution, or financial-information provision that affect the salience or perception of relative economic position. By changing the information people have about others, such policies can alter perceived relative standing even without changing individuals' economic resources. Relatedly, political actors and interest groups frequently attempt to shape these perceptions to gain support for specific agendas. Our results suggest that the effects of these policies on risk-taking may therefore depend on the distribution of control beliefs in the population. While policies generally cannot be tailored to individuals' locus of control, accounting for this heterogeneity may be important for anticipating their behavioral and distributional consequences.

Like most empirical case studies, our study is not without limitations. To elicit risk-taking patterns in an incentivized setting, we had to rely on lottery choices with moderate stakes. A natural question is thus how our results carry over to higher-stakes settings. Relatedly, our focus is risk-taking in the gain domain. Negative outcomes are, however, common in many real-world decisions. For example, insurance choices provide a natural setting for examining whether our findings extend to the loss domain. While our focus on risk-taking helps us predict behavioral responses to relative wealth across various domains and real-world settings, testing these predictions in these settings directly remains an important avenue for future research.

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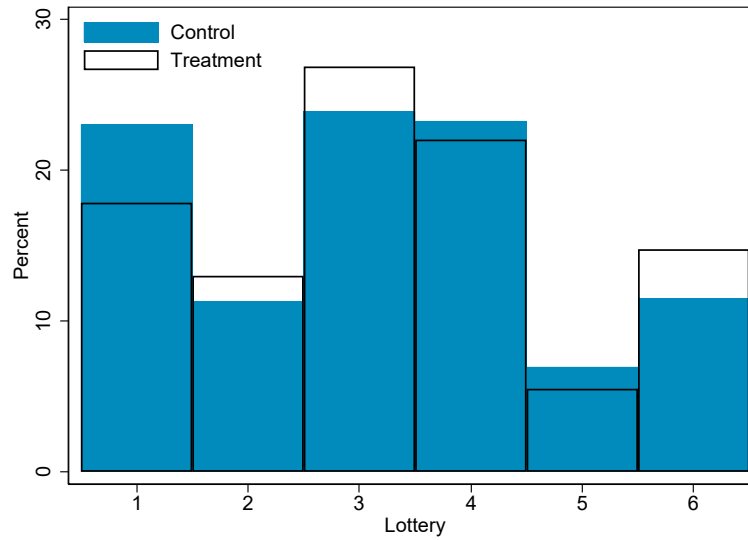
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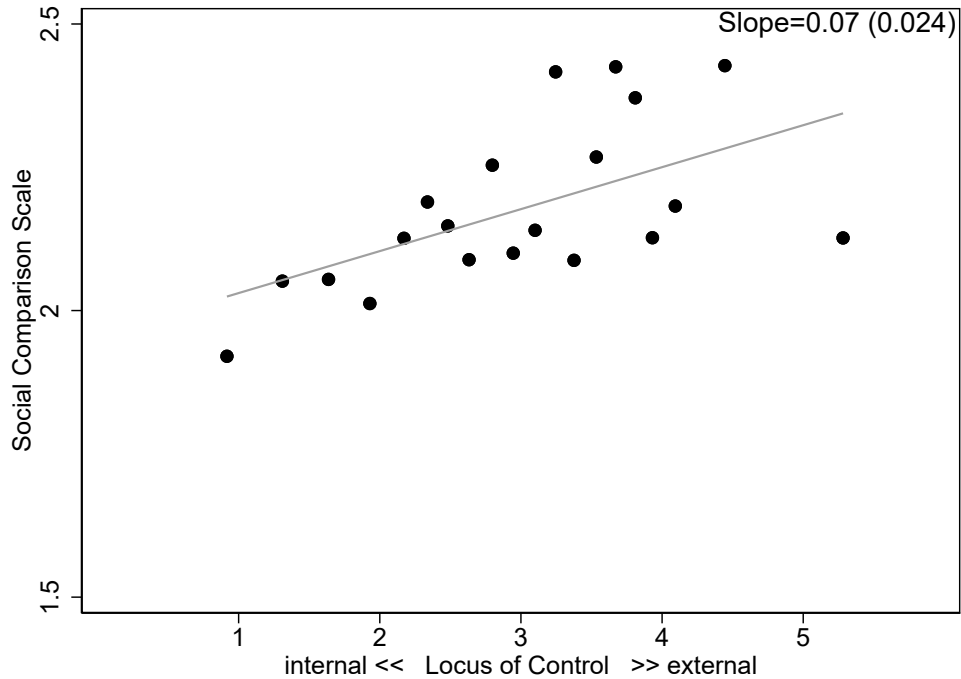
Figures

Figure 1: Lottery Choices by Treatment



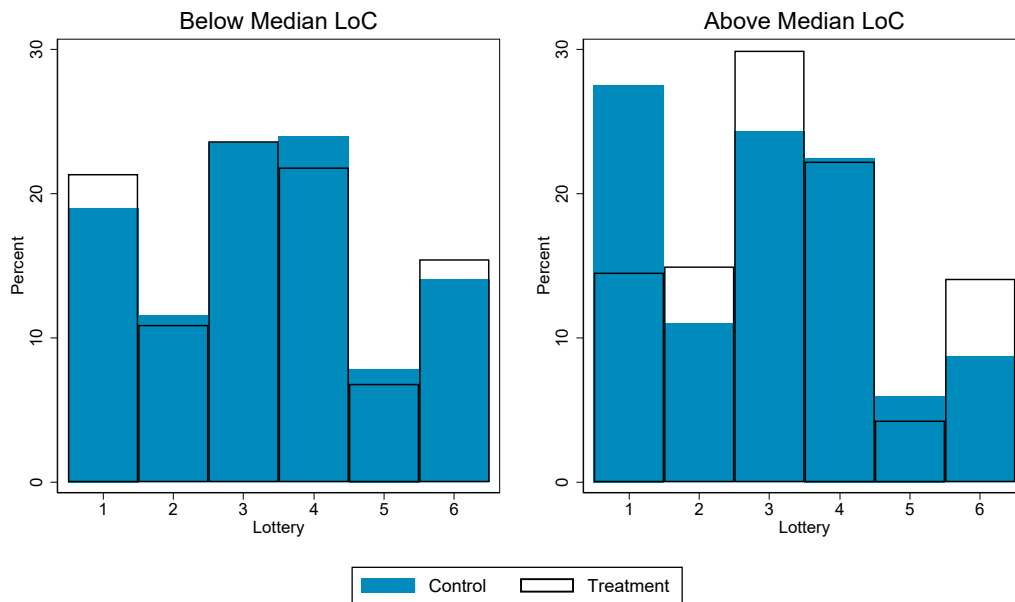
Notes: Distribution of lottery choices in the control (blue bars) and treatment condition (white bars). Lottery 1 pays €50 with equal probability (safe lottery), and subsequent lotteries become more risky, with lottery 6 paying €200 and €0 with equal probability. See Table [A2](#) for details.

Figure 2: Locus of Control and Relative Comparisons



Notes: Binned scatterplots of LoC and relative comparisons using data from a different representative SOEP-IS sample ($N = 1,388$). LoC is proxied by the beliefs that *life is mostly controlled by fate* and that *working hard enough likely results in a good life*; we constructed an equally weighted index of these two beliefs, with higher values indicating greater external belief. Relative comparisons are measured using a short version of the Social Comparison Scale (Schneider and Schupp, 2011), an equally weighted index of three statements about a person's tendency to engage in social comparisons, with higher values indicating greater disposition towards social comparison. The statements are *I often compare myself with others with respect to what I have accomplished in life*, *I often compare my social skills and popularity with those of other people*, and *I always pay very strong attention to how I do things compared to others*. All measured on a scale from 1 (disagree strongly) to 5 (agree strongly). Control variables include age, gender, education, parents' education, equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), baseline risk aversion, homeownership, satisfaction with health, life satisfaction, and region (East/West Germany). Data: SOEP-IS Sample I2/I4.

Figure 3: Lottery Choices by Treatment and LoC



Notes: Distribution of lottery choices in the control (blue bars) and treatment condition (white bars), separate by below (internal, left panel) and above median LoC (external, right panel). Lottery 1 pays €50 with equal probability (safe lottery), and subsequent lotteries become more risky, with lottery 6 paying €200 and €0 with equal probability. See Table [A2](#) for details.

Tables

Table 1: Wealth Categories by Treatment

Control Group		Treatment Group	
<i>Wealth Category (in €)</i>	<i>% responses</i>	<i>Wealth Category (in €)</i>	<i>% responses</i>
<2,500	27.05	<275,000	79.01
2,501 – 11,000	20.00	275,001 – 468,000	12.74
11,001 – 27,000	11.59	468,001 – 722,000	5.19
27,001 – 112,000	16.82	722,001 – 989,000	1.65
>112,000	24.55	>989,000	1.42

Notes: Wealth categories used in the two conditions and share of observed responses in each category. Upper bounds taken from the 2014 Household Finance and Consumption Survey (HFCS, Deutsche Bundesbank, 2016). Data: SOEP-IS Sample I3.

Table 2: First Stage Effects of Treatment

	Abs. Wealth	Adequacy	Assets	Q. Rel. Wealth	Rel. Wealth	Top 10%
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	-0.088 (0.057)	-0.017 (0.060)	-0.002 (0.070)	-0.175*** (0.048)	-0.170*** (0.062)	0.307*** (0.110)
Sample Observations	Bilendi 900	Bilendi 900	Bilendi 900	respondi 987	respondi 987	SOEP-IS 865

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. OLS regression (columns 1–5) and GLM regressions (gamma-log; column 6) with robust standard errors in parentheses. “Abs. Wealth” is a respondent’s perceived absolute wealth situation (“If you consider only your own household’s situation, how would you describe your household’s current net wealth? (1) Very low – (5) Very high”), “Adequacy” is a respondent’s perceived adequacy of their absolute wealth level (“If you consider only your own household’s situation, how adequate is your household’s current net wealth for your household’s needs and plans?” (1) Not at all adequate – (5) Fully adequate), and “Assets” is the number of the mentioned risky asset classes plus homeownership (“Which of the following assets (your own or those of other household members) did you include in your wealth estimate?” – asset list and multiple answers possible). “Q. Rel. Wealth” is a respondent’s perceived wealth situation relative to other households (“Compared to most other households in Germany, would you say that your wealth situation ...? is much worse, worse, the same, is better, or much better”). “Rel. wealth” is a respondent’s perceived rank in the wealth distribution (“What do you think is the share of households in Germany that have less net wealth than your household?”). All are standardized to zero mean and unit variance. “Top 10%” is the estimated threshold for being in the top 10% of the pre-tax household income distribution in Germany (“In your estimation, what gross annual income do you need to be in the top 10 percent of German households?”). Covariates include age, gender, education, parents’ education, (equivalized) net income, marital status, household size, employment status (self-employed, unemployed, retired), homeownership, and region (East/West Germany). Data: respondi, Bilendi and SOEP-IS Sample I3.

Table 3: Average Treatment Effect on Risk Taking

	= 1, if Lottery 1 (1)	Lottery Choice (Cat.) (2)	CRRRA (3)
Treated	−0.056** (0.027)	0.121* (0.070)	−0.584** (0.279)
Control Group Mean	0.230		2.863
Model	LPM	Ordered Probit	Interval
Observations	914	914	914

Notes: * ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). Column (1) presents the estimate from a linear probability model, in which the outcome is an indicator for choosing Lottery 1 (€50 payoff). Column (2) reports the point estimate from an ordered probit model, where the outcome is a categorical variable indicating all six lottery choice options; for the marginal effects on each lottery choice, see Table A6. Column (3) reports the estimate from an interval regression, where the outcome is the CRRRA parameter ρ (higher values indicate more risk-averse behavior). All regressions adjust for age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), and the Big 5 personality traits. Standard errors in parentheses. Sample: SOEP-IS Sample I3.

Table 4: Treatment Effect on Risk Taking; by LoC Beliefs

	= 1, if Lottery 1			CRRA		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.018 (0.038)	-0.136*** (0.039)	-0.054** (0.027)	0.080 (0.401)	-1.387*** (0.386)	-0.564** (0.278)
Treated $\times$ LoC			-0.095*** (0.028)			-0.914*** (0.280)
LoC			0.035 (0.022)			0.335 (0.219)
Sample	Int. LoC	Ext. LoC	Full	Int. LoC	Ext. LoC	Full
Model	LPM	LPM	LPM	Interval	Interval	Interval
Observations	462	452	914	462	452	914

Notes: * ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). Columns (1)-(3) present estimates from linear probability models, in which the outcome is an indicator for choosing Lottery 1 (€50 payoff). Columns (4)-(6) report the estimates from interval regressions, where the outcome is the CRRA parameter ρ (higher values indicate more risk-averse behavior). All regressions adjust for age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), and the Big 5 personality traits. Standard errors in parentheses. Sample: SOEP-IS Sample I3.

Appendix – For Online Publication Only

Status, Control, and Risk: How Relative Wealth Perceptions Shape Risk-Taking

Dietmar Fehr and Yannick Reichlin

A.1 Manipulation Checks

Setup. We conducted two separate survey experiments to examine whether the manipulation had the intended effect of inducing a feeling of low relative wealth. This allowed us to economize on the limited survey time in SOEP-IS, where we only implemented a basic test of how the intervention affects relative standing, and to conduct comprehensive manipulation checks without risking contamination of the outcomes by the first stage, or of the first stage by the outcomes or the manipulation.

We recruited a new sample ($N = 987$) through *respondi* and another ($N = 900$) through *Bilendi*, two internationally well-known panel providers, since we no longer had access to the SOEP-IS. In each sample, we implemented a short survey (median duration: 6 minutes), in which we randomized respondents to treatment and control conditions using the same intervention as in the initial survey. That is, we either presented five wealth categories with wide intervals (treatment) or small intervals (control), and respondents had to indicate into which category their net wealth falls. Note that we updated the upper bounds of the intervals according to the latest available wealth data for Germany (see Table A19).

Manipulation check (respondi). The aim of the first manipulation check was to directly measure the impact of the intervention on (i) respondents' perceived relative position in the wealth distribution and (ii) the perceived wealth distribution benchmarks. We asked how respondents perceived their wealth situation relative to other households and how they perceived their relative rank in the German wealth distribution.

1. Compared to most other households in Germany, would you say that your wealth situation is ...? [*much worse, worse, the same, better, much better*]

2. What do you think is the share of households in Germany that have less net wealth than your household?

0 percent (%) means that all households have more net wealth than your household and 100 percent (%) means that no household has more net wealth.

Next, we examined respondents' perceived wealth distribution benchmarks and elicited the perceived net wealth at the 50th and 90th percentiles of the net wealth distribution. To familiarize respondents with this rather involved task, we first explained the meaning of wealth distribution, how to partition the distribution into 10 equally sized bins, and what the median and 90th percentile are. To ease understanding, we illustrated this graphically with a ladder of nine rungs that represents the bottom 10%, the median, and the top 10% households. Subsequently, respondents estimated the net wealth of the median household and the top 10% household using a slider displaying values according to the function $Y = 500 * \exp^{(0.045 * steps)}$, where "steps" correspond to the slider range from 0 to 200.¹ The specific questions are:

Now suppose we rank all households in Germany by net wealth from lowest to highest and divide this ranking into 10 equal sections, as shown on the ladder. On the ladder, the 10% poorest households would then be on the bottom rung and the 10% richest households on the top rung. The household in the middle divides the ranking into two halves, i.e. 50% of the households are richer, and 50% are poorer than the household in the middle.

3. What do you think is the net wealth of the household in the middle of the ladder?
4. You just estimated that the household in the middle of the ladder has a net wealth of *[estimated median]*€.

What do you think is the required net wealth to belong to the 10% richest households (i.e. to be on the top rung of the ladder)?

The survey concluded with a few questions on socio-economic characteristics and attitudes, including locus of control (LoC), self-esteem, and income adequacy.

Manipulation check (Bilendi). In the second survey, we aimed to measure the impact of the intervention on (i) respondents' perceived absolute wealth, (ii) emotional states, and (iii) experimenter demand. We asked the following questions after the treatment:

¹Due to a programming error, the slider range of the top 10% question started with a respondent's median wealth estimate, such that each respondent faced a different scale. Therefore, we do not use this question in our analysis.

1. Which of the following assets (your own or those of other household members) did you include in your wealth estimate? *[asset list and multiple answers possible]*
2. If you consider only your own household's situation, how would you describe your household's current net wealth? *(1) Very low – (5) Very high*
3. If you consider only your own household's situation, how adequate is your household's current net wealth for your household's needs and plans? *(1) Not at all adequate – (5) Fully adequate*
4. Please think about the question regarding your household's net wealth and indicate how strongly you are currently feeling the following emotions. Do you feel . . . ? *[happy, angry, sad, worried] (1) Not at all – (5) Very strongly*

In addition, we elicited respondents' socio-economic characteristics and attitudes and asked them to guess the purpose of the survey (open-ended question) and to indicate whether anything caught their attention or might have influenced their responses during the survey and, if so, what it was (open-ended question).

Results. We discuss the main results from this manipulation check in detail in Sections 3 and 4.3 and briefly summarize additional results here.

In the first manipulation check (respondi) we ask respondents to estimate the wealth at the 50th and 90th percentiles of the wealth distribution. At the 50th-percentile benchmark, treated respondents estimate median wealth higher than non-treated respondents, but the difference is insignificant (coeff. 0.204, s.e. 0.158). We do not look at the 90th percentile due to the programming error mentioned above.

In the second manipulation (Bilendi), we address a potential concern that the asymmetric wealth categories, especially in the treatment condition, capture respondents' attention. Specifically, we asked respondents the following question: "Was there anything during the survey that particularly caught your attention or that you thought might influence your answers?" (Yes/No/Don't know – if they answered "yes," they could specify it in an open-text field). About five percent (46 out of 900 respondents) answered "Yes." Of these respondents, only 22 mentioned that the "question about wealth" and/or the "categories" caught their attention, whereas 24 mentioned other things. This suggests that demand effects are unlikely to be influencing our results. Note also that the original SOEP-IS survey is more than an hour long and covers a wide range of topics, whereas the manipulation check focuses primarily on wealth. This contrast may increase the

salience of certain wealth details in the manipulation check, suggesting that the five percent may be seen as an upper bound.

A.2 Income Inequality, Risk-Taking, and Control Beliefs

In this section, we document the relationship between income inequality and risk-taking, and that an external locus of control moderates this relationship. Inequality is tightly linked to the relative position in the wealth (and income) distribution. The distribution of wealth affects how much richer others are, and this notion of relative deprivation translates into the Gini coefficient at the aggregate level (Yitzhaki, 1979; Hey and Lambert, 1980). The reported findings are based on the analysis of data from two large-scale international surveys: the Global Preference Survey (Falk et al., 2018) and the World Values Survey (Inglehart et al., 2014).

Global Preference Survey (GPS). The GPS contains a validated risk measure, elicited in nationally representative samples of 76 countries spanning some 90% of the world’s population. The risk measure is a weighted average of a qualitative self-assessment of risk tolerance akin to the qualitative measure in the SOEP (see section 4.3) and a quantitative measure of a sequence of lottery choices. We focus here on the aggregated level and compare national-level summary statistics of the risk measure with a national-level measure of inequality, the Gini index, taken from the World Bank’s World Development Indicators (WDI). The two measures are significantly correlated ($p=0.005$) with a Pearson correlation coefficient of 0.33, suggesting that in more unequal countries, the population is, on average, more risk-tolerant. Pickard et al. (2024), in work initiated after our initial working paper, provide a more detailed account of inequality and risk-taking using the GPS. We next turn to an individual-level analysis to examine the moderating effect of LoC.

World Values Survey (WVS). The WVS collects socio-demographic information, political attitudes, and value judgments from nationally representative samples in repeated cross-sections. Wave 5 (2005-2009) and Wave 6 (2010-2014) contain two questions that are particularly relevant to our purposes. First, the waves include a measure of risk attitudes developed by Schwartz (1992), which asks respondents to compare themselves to a hypothetical person who finds it important to “experience adventure and [take] risks.” Answers range on a scale from (1) *very much like me* to (6) *not at all like me*. The measure captures sensation-seeking as developed by personality psychologists and is conceptually related to economists’ notion of risk-aversion (Almlund et al., 2011). Second,

they include a proxy for locus of control. Respondents are asked about how much freedom of choice and control they feel they have over the way their life turns out on a scale from (1) *none at all* to (10) *a great deal*. To ease interpretation, we standardize answers to both questions by their sample mean and standard deviation and recode the LoC question such that higher values imply more external beliefs.

In our analysis, we use Waves 5 and 6 of the WVS and merge this data with information on national-level inequality from the WDI. This leaves us with a sample of 145,206 individual observations, coming from 71 countries and spanning 12 years. We estimate linear regressions of measured risk attitudes on a proxy for inequality, locus-of-control beliefs, and their interaction. Table A18 shows that individual risk aversion is negatively correlated with national-level inequality. This complements the aggregate-level correlation we have seen in the GPS that is based on a validated risk measure. Moreover, we see that this relationship is considerably more pronounced for individuals with external control beliefs and is robust to the inclusion of a battery of observable socio-demographic variables, survey-year fixed effects, and dummies for cultural country clusters.

This result from the WVS suggests that our evidence has broader significance, as it applies to the distribution of income, which itself may affect the extent to which people seek relative comparisons, and that it extends beyond the German context.

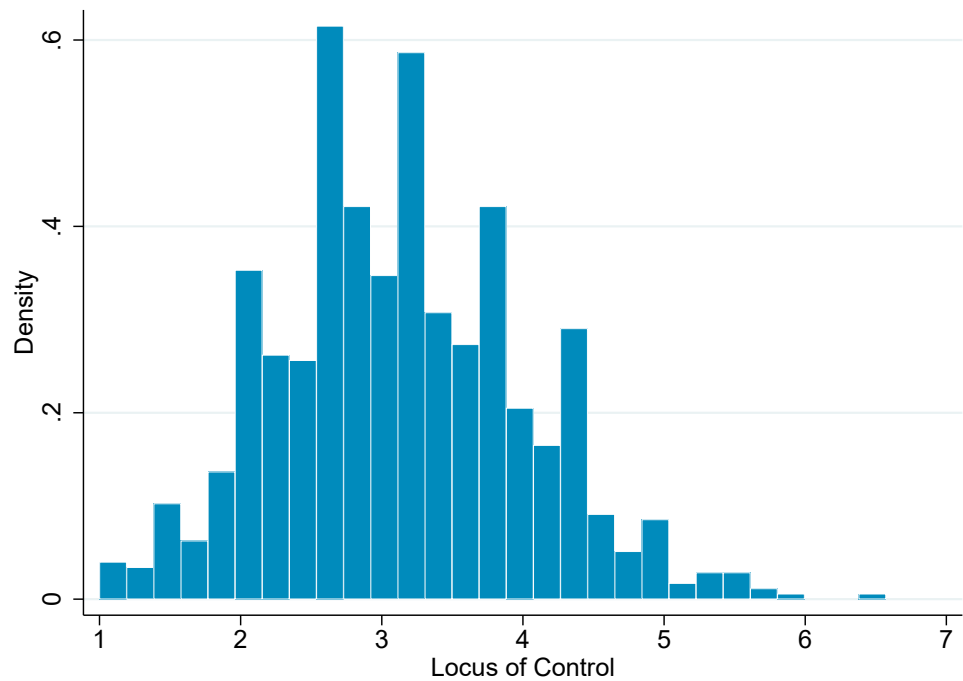
A.3 Multiple Hypothesis Test Corrections and Randomization Inference

In Table 4, we show that respondents with more external beliefs react more strongly to our treatment. To test the robustness of this result, we examine a host of other personality traits (Table A13), emotions (Table A15), and socio-economic factors (Table A16) that may potentially interact with the treatment in some meaningful way. In total, we test 15 variations. Thus, we control for multiple hypothesis testing using the false discovery rate (FDR), which is the expected proportion of false rejections among all rejected null hypotheses. Following Benjamini et al. (2006), we apply the two-stage linear step-up procedure to control for the FDR. Among the 15 hypotheses, only our main result on LoC is significant at a conventional level ($p = 0.001$, Table 4, column 6). This p-value is smaller than the threshold $q^* = 0.0034$ derived from the two-stage linear step-up procedure, the threshold of the Holm correction $\tilde{p} = 0.0033$, and the Bonferroni correction $\tilde{p} = 0.0033$. Accordingly, we can reject the null hypothesis of no interaction between LoC and treatment at both the conventional and adjusted p-values.

Similarly, when testing the interaction between LoC and our treatment using a randomization inference approach, we observe that only 4 out of 5,000 generated interactions (0.08%) are as large or larger than the one observed in our experiment ($p < 0.001$).

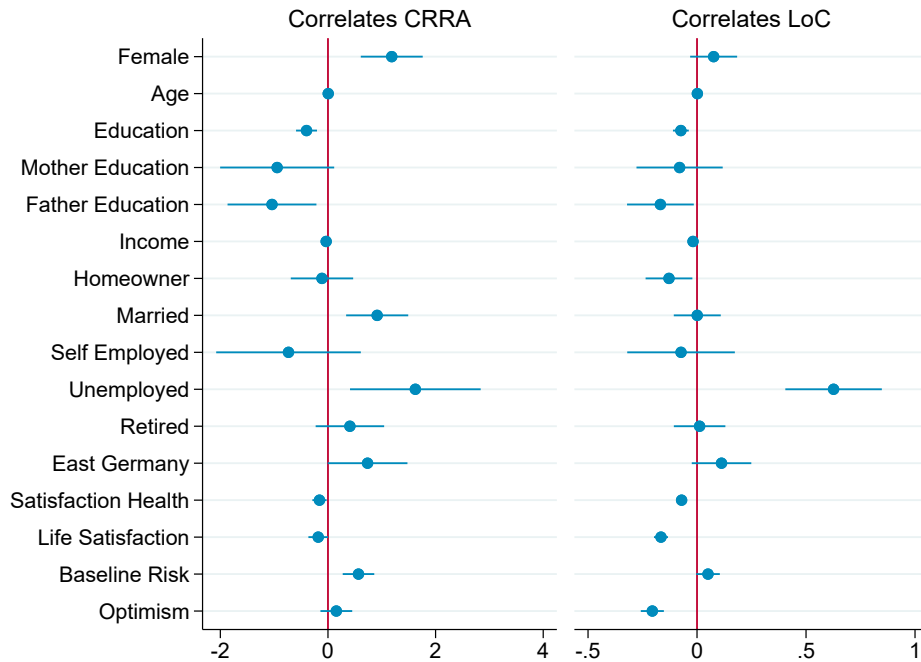
A.4 Additional Figures

Figure A1: Distribution of Locus of Control



Notes: Histogram of Locus of Control (LoC) based on seven equally weighted items of the LoC questions (see Table A1 for the items and details on the construction of the index). Higher values imply more external control beliefs. Data: SOEP-IS Sample I3.

Figure A2: Correlates of Risk and Locus of Control



Notes: Coefficient plots from separate OLS regressions of the form $Y_i = \beta_0 + \beta_1 \cdot \text{Covariate} + \epsilon_i$, where Y_i is either the CRRA parameter ρ (with higher values indicating less tolerance for risk) or Locus of Control (with higher values indicating more external control beliefs; see Table A1 for more details on the construction of the index). *Education* is measured in 7 categories according to the International Standard Classification of Education (ISCED), with higher categories representing a higher level of education. *Mother (Father) has Abitur* indicates a parent with qualification for university admission (Abitur is the final exam at the end of secondary school). *Income* is the equivalence scale determined by the square root scale (i.e., we divide the monthly household net income by the square root of the number of household members). *East Germany* is an indicator for respondents who lived in East Germany before 1990. *Homeowner* identifies respondents who own their house or apartment. *Female*, *Married*, *Self-employed*, *Unemployed*, and *Retired* are indicator variables. *Satisfaction with health* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Life Satisfaction* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Baseline Risk* is a qualitative risk measure on a scale from 0 (very willing to take risks) to 10 (not at all willing to take risks), standardized using the sample mean and standard deviation. *Optimism* is measured on a scale from 1 (optimistic) to 4 (pessimistic) and recoded such that higher values reflect more optimism. Data: SOEP-IS Sample I3.

A.5 Additional Tables

Table A1: Locus of Control Questions

Locus of Control Module Items					
	N	Mean	S.D.	Min	Max
IT1: The course of my life depends on me (<i>reverse coded</i>).	914	2.36	1.31	1	7
IT2: In comparison to others, I have not achieved what I deserve.	914	3.11	1.70	1	7
IT3: What you achieve in life is first of all a question of fate or luck.	914	3.49	1.61	1	7
IT4: I often experience that others are deciding about my life.	914	2.70	1.58	1	7
IT5: You have to work hard to be successful (<i>reverse coded</i>).	914	2.09	1.12	1	7
IT6: When I face difficulties in life, I often doubt my abilities.	914	3.17	1.62	1	7
IT7: Which opportunities I have in life is determined by social conditions.	914	4.45	1.52	1	7
IT8: More important than all effort are the abilities you have.	914	4.83	1.35	1	7
IT9: I have little control over the things happening in my life.	914	2.66	1.43	1	7
IT10: Social or political involvement can influence social conditions.	914	3.96	1.65	1	7
Locus of Control Indices					
$LoC = (IT1+IT2+IT3+IT4+IT6+IT7+IT9)/7$	914	3.13	0.87	1	6.57
<i>External Scale</i> (IT2 IT3, IT4, IT6, IT7, IT9)	914	3.09	0.99	1	6.67
<i>Internal Scale</i> (IT1, IT5, IT8)	914	3.05	0.68	1	5.26
<i>Overall LoC</i> (IT1, IT2, IT3, IT4, IT5, IT6, IT7, IT8, IT9)	914	3.06	0.94	1.02	6.38

Notes: Descriptive statistics for each item in the Locus of Control (LoC) module. Each item required respondents to answer on a scale from 1 (disagree completely) to 7 (Agree completely). Items were coded such that higher values reflect more external control beliefs. We follow the SOEP scales manual Richter et al. (2013) and Specht et al. (2013) and combine *IT1, IT2, IT3, IT4, IT6, IT7, IT9* in an equally weighted *LoC* index as in Cobb-Clark and Schurer (2013). These items have the highest reliability (Cronbach's $\alpha = 0.66$). As a robustness check, we consider two separate indices for internal and external control beliefs and a weighted single index (*Overall LoC*) including all items loading on internal or external locus of control (Cronbach's $\alpha = 0.59$). We construct two separate indices using principal component factor analysis to identify which items load on internal and external locus of control, then estimate single-factor loadings for each and build standardized weighted indices from those loadings, following Caliendo et al. (2015); Piatek and Pinger (2016); Fitzenberger et al. (2021); Caliendo et al. (2022). The weights for the single index (*Overall LoC*) are determined by a factor analysis on all nine items used in the two separate indices. For the regression analyses, we standardize all indices to a mean of zero and a standard deviation of one. Data: SOEP-IS Sample I3.

Table A2: Overview of Gambles

	Payoffs	EV	S.D.	CRRA-Interval
<i>Lottery 1</i>	(50, 50)	50	0	$[7.51, \infty)$
<i>Lottery 2</i>	(45, 95)	70	25	$[1.74, 7.51]$
<i>Lottery 3</i>	(40, 120)	80	40	$[0.812, 1.74]$
<i>Lottery 4</i>	(30, 150)	90	60	$[0.315, 0.812]$
<i>Lottery 5</i>	(10, 190)	100	90	$[0, 0.315]$
<i>Lottery 6</i>	(0, 200)	100	100	$(-\infty, 0]$

Notes: Lotteries used in the risk elicitation task. Each lottery pays a low or high payoff (in €) with equal probability. Respondents choose one of the lotteries, and for every tenth respondent, the chosen lottery was realized and paid out. Assuming CRRA utility of the form $u(x) = \frac{x^{1-\rho}}{1-\rho}$, where x is the respective payoff and ρ the coefficient of interest, we can calculate lower and upper bounds for relative risk aversion for each lottery. For example, comparing the expected utility of Lottery 2 and Lottery 3 gives the lower bound of ρ for Lottery 2 and the upper bound of ρ for Lottery 3.

Table A3: Balance

	Mean Treatment	Mean Control	p-value
Female=1	0.54	0.53	0.906
Age	48.27	46.63	0.170
Lower secondary=1	0.15	0.14	0.915
Upper secondary=1	0.63	0.64	0.743
College=1	0.23	0.22	0.772
Mother has Abitur=1	0.08	0.11	0.126
Father has Abitur=1	0.14	0.17	0.319
Income (net/month)	2075.35	2203.91	0.076
Home owner=1	0.49	0.48	0.736
Log(wealth)	10.49	10.10	0.492
Married=1	0.53	0.54	0.633
No. of HH Members	2.35	2.44	0.218
Self employed=1	0.05	0.04	0.557
Unemployed=1	0.06	0.05	0.327
Retired=1	0.29	0.26	0.199
East Germany=1	0.19	0.17	0.303
Satisfaction with Health	6.79	6.62	0.273
Life Satisfaction	7.61	7.62	0.922
Baseline Risk	5.04	5.11	0.643
<i>Prob > F</i>			0.59

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row presents the means of *covariate* in the treatment and control group along with the p-values from separate OLS regressions of the form $Treated = \beta_0 + \beta_1 * Covariate + \epsilon_i$. *Prob > F* is the p-value from an F-test for joint significance of all covariates. Education is measured in 7 categories according to the International Standard Classification of Education (ISCED), where *Lower secondary* corresponds to the first two categories, *Upper secondary* to categories 3, 4, and 5, and *College* to the highest category (6). *Mother (Father) has Abitur* indicates a parent with a qualification for university admission (Abitur is the final exam at the end of high school). *Income* is the equivalence scale determined by the square root scale (i.e., we divide the monthly household net income by the square root of the number of household members), *Homeowner* identifies respondents who own their house or apartment, and *Log(wealth)* is the log of the estimated wealth level of the household. *Female*, *Married*, *Self-employed*, *Unemployed*, and *Retired* are indicator variables. *East Germany* is an indicator for respondents who lived in East Germany before 1990. *Satisfaction with health* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Life Satisfaction* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Baseline Risk* is a qualitative risk measure on a scale from 0 (very willing to take risks) to 10 (not at all willing to take risks). Data: SOEP-IS Sample I3.

Table A4: Personality Traits – Balance

	Mean Treatment	Mean Control	p-value
Openness	-0.00	0.02	0.762
Conscientiousness	-0.02	-0.01	0.889
Extraversion	0.01	0.03	0.730
Agreeableness	-0.02	-0.00	0.820
Neuroticism	0.00	0.01	0.888
Optimism	-0.01	0.05	0.343
Relative Optimism	0.01	-0.01	0.776
Locus of Control	3.10	3.17	0.257
Self Control	-0.04	-0.06	0.773
<i>Prob > F</i>			0.964

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row presents the means of *covariate* in the treatment and control group along with the p-values from separate OLS regressions of the form $Treated = \beta_0 + \beta_1 * Covariate + \epsilon_i$. *Prob > F* is the p-value from an F-test for joint significance of all covariates. The Big 5 consist of *Openness*, *Conscientiousness*, *Extraversion*, *Agreeableness*, and *Neuroticism*, which are standardized to a mean of zero and a standard deviation of 1. *Optimism* indicates how respondents think about their future (“If you think about the future: Are you... (1) optimistic; (2) rather optimistic than pessimistic; (3) rather pessimistic than optimistic; (4) pessimistic?”). We recoded the variable such that higher values reflect more optimism. *Relative optimism* is the unweighted average of four questions for which respondents had to indicate on a scale from 1 (“Very much less likely”) to 7 (“Very much more likely”) how they judged the likelihood to: (1) being financially successful; (2) not becoming seriously ill; (3) being successful in their job; (4) being happy in general, relative to peers of the same age and sex. *Locus of Control (LoC)* is an equally weighted index of the LoC questions, with higher values corresponding to more external beliefs. *Self-Control* is a standardized (mean zero, standard deviation 1) sum of 13 self-control indicators, in the spirit of Tangney et al. (2004). Data: SOEP-IS Sample I3.

Table A5: Manipulation Check: Credibility, LoC and Emotions

	Credibility	LoC	Self-esteem	Pos. Affect	Neg. Affect	Affect Balance
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	0.006 (0.063)	-0.055 (0.063)	0.025 (0.061)	-0.063 (0.064)	0.071 (0.065)	-0.081 (0.063)
Sample	respondi	respondi	respondi	Bilendi	Bilendi	Bilendi
Observations	987	987	987	900	900	900
R^2	0.07	0.08	0.10	0.13	0.10	0.14

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. OLS regression with robust standard errors in parentheses. “Credibility” measures the credibility of the data on wealth categories (“The data on wealth categories comes from the Bundesbank. How credible do you consider these data to be?”; scale: 1 – not credible to 5 – very credible), “LoC” is an index for internal and external locus of control measured on a scale from 1 (internal) to 7 (external), “Self-esteem” measures a positive self-attitude (“I have a positive attitude towards myself.”; scale: 1 – does not apply at all to 7 – does apply), and “Pos. Affect” and “Neg. Affect” are measures for emotional responses (happiness, anger, worry, sadness) to the treatment (on a scale from 1 to 5). “Affect Balance” is the difference between positive and negative affect, with higher values indicating a more positive balance. All measures are standardized. Covariates include age, gender, education, parents’ education, equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), homeownership, and region (East/West Germany). Data: respondi and Bilendi.

Table A6: Average Marginal Effects of Treatment on Lottery Choice

	Change in probability of choosing each lottery		
Lottery 1	−0.033* (0.019)	−0.003 (0.027)	−0.082*** (0.028)
Lottery 2	−0.009* (0.005)	−0.001 (0.007)	−0.022*** (0.008)
Lottery 3	−0.004 (0.002)	−0.000 (0.004)	−0.005 (0.004)
Lottery 4	0.014* (0.008)	0.001 (0.010)	0.038*** (0.013)
Lottery 5	0.007* (0.004)	0.001 (0.006)	0.016*** (0.006)
Lottery 6	0.025* (0.015)	0.002 (0.022)	0.056*** (0.019)
Observations	914	462	452
Sample	All participants	Below-median LoC	Above-median LoC

Notes: Entries are average marginal effects of treatment from ordered probit models, expressed as percentage-point changes in the probability of choosing each lottery. Positive values indicate that treatment increases the probability of choosing the corresponding lottery. The first column reports estimates for the full sample. The second and third columns report estimates for participants with below- and above-median locus-of-control (LoC) scores, respectively. Higher LoC scores indicate more external control beliefs. All models include the controls listed in Table 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: SOEP-IS Sample I3.

Table A7: Correlates of Lottery Choice, Risk Aversion, and Locus of Control

	Lottery Choice		CRRA Parameter		Locus of Control	
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.450*** (0.149)	-0.339** (0.149)	1.185*** (0.415)	0.911** (0.409)	0.076 (0.078)	0.042 (0.073)
Age	0.001 (0.004)	0.005 (0.006)	0.005 (0.011)	-0.007 (0.017)	0.002 (0.002)	0.002 (0.003)
Education	0.169*** (0.050)	0.119** (0.056)	-0.399*** (0.140)	-0.316** (0.153)	-0.074*** (0.026)	-0.044 (0.028)
Mother has Abitur	0.226 (0.275)	-0.233 (0.312)	-0.944 (0.765)	0.423 (0.859)	-0.080 (0.143)	0.115 (0.154)
Father has Abitur	0.363* (0.213)	0.288 (0.236)	-1.040* (0.596)	-0.704 (0.650)	-0.168 (0.111)	-0.066 (0.116)
Income (net/month)	0.013* (0.007)	0.001 (0.008)	-0.033* (0.020)	-0.007 (0.023)	-0.018*** (0.004)	-0.011*** (0.004)
Wealth(log)	0.007 (0.009)	0.003 (0.010)	-0.014 (0.024)	-0.009 (0.027)	-0.012*** (0.004)	-0.003 (0.005)
Married	-0.173 (0.150)	-0.345** (0.159)	0.914** (0.416)	1.336*** (0.439)	0.001 (0.078)	0.148* (0.078)
Self employed	0.444 (0.343)	0.075 (0.355)	-0.733 (0.969)	0.262 (0.982)	-0.073 (0.178)	0.091 (0.175)
Unemployed	-0.761** (0.311)	-0.583* (0.338)	1.624* (0.876)	1.183 (0.934)	0.626*** (0.160)	0.263 (0.166)
Retired	-0.109 (0.164)	-0.211 (0.245)	0.408 (0.459)	0.375 (0.679)	0.012 (0.085)	-0.027 (0.120)
East Germany	-0.121 (0.190)	-0.118 (0.192)	0.736 (0.535)	0.783 (0.533)	0.112 (0.099)	0.001 (0.095)
Satisfaction with Health	0.063* (0.034)	0.022 (0.040)	-0.157* (0.095)	-0.038 (0.110)	-0.071*** (0.017)	0.005 (0.020)
Life Satisfaction	0.085* (0.047)	0.066 (0.054)	-0.181 (0.132)	-0.153 (0.150)	-0.165*** (0.023)	-0.122*** (0.027)
Baseline Risk	-0.195*** (0.075)	-0.187** (0.075)	0.567*** (0.211)	0.564*** (0.210)	0.050 (0.039)	0.029 (0.037)
Optimism	-0.059 (0.076)	-0.115 (0.079)	0.155 (0.213)	0.308 (0.217)	-0.205*** (0.038)	-0.136*** (0.039)
Observations	460		460		460	

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the lottery choice in columns 1-2, the CRRA parameter ρ (with higher values indicating less tolerance for risk) in columns 3-4, and Locus of Control (with higher values indicating external control beliefs) in columns 5-6. Data from the control group only; interval regressions in columns 3-4 and otherwise OLS regressions (with standard errors in parentheses). Odd-numbered columns display coefficients from separate regressions for each covariate, while even-numbered columns report a multivariate regression including all covariates at once. *Education* is measured in 7 categories according to the International Standard Classification of Education (ISCED), with higher categories representing a higher level of education. *Mother (Father) has Abitur* indicates a parent with a qualification for university admission (Abitur is the final exam at the end of high school). *Income* is the equivalence scale determined by the square root scale (i.e., we divide the monthly household net income by the square root of the number of household members) and *Homeowner* identifies respondents who own their house or apartment. *Female*, *Married*, *Self-employed*, *Unemployed*, and *Retired* are indicator variables. *East Germany* is an indicator for respondents who lived in East Germany before 1990. *Satisfaction with health* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Life Satisfaction* is measured on a scale from 0 (completely dissatisfied) to 10 (completely satisfied). *Baseline Risk* is a qualitative risk measure on a scale from 0 (very willing to take risks) to 10 (not at all willing to take risks), standardized using the sample mean and standard deviation. *Optimism* is measured on a scale from 1 (optimistic) to 4 (pessimistic) and recoded such that higher values reflect more optimism. Data: SOEP-IS Sample I3.

Table A8: Heterogeneous Effects - Locus of Control; Ordered Probit

	(1)	(2)	(3)
Treated	0.010 (0.101)	0.303*** (0.102)	0.119* (0.071)
Treated $\times$ LoC			0.182** (0.071)
LoC			-0.076 (0.056)
Sample	Below Median	Above Median	Full
Observations	462	452	914

Notes: * ($p < 0.1$), ** ($p < 0.05$), *** ($p < 0.01$). The table reports point estimates from ordered probit models, where the outcome is a categorical variable indicating all six lottery choice options; for the predicted marginal effects on each lottery choice, see Table A6. Columns (1) and (2) are estimated on the sample of participants with below- and above-median LoC beliefs, respectively. Column (3) is based on the full sample. LoC is the z-score of a single index for locus of control, constructed as detailed in Table A1. Higher values correspond to more external control beliefs. All regressions adjust the same set of covariates as Table 3. Standard errors in parentheses. Sample: SOEP-IS Sample I3.

Table A9: Heterogeneous Effects - Locus of Control; Functional Form Robustness

	CRRA Parameter			
	(1)	(2)	(3)	(4)
Treated	-0.564** (0.278)	-0.851** (0.343)	-0.851** (0.350)	-0.694** (0.308)
Treated x LoC	-0.914*** (0.280)	-0.997*** (0.286)	-1.004** (0.438)	-0.806* (0.446)
Treated x LoC ²		0.300 (0.203)	0.299 (0.227)	
Treated x LoC ³			0.003 (0.125)	
Sample Observations	Full 914	Full 914	Full 914	Drop Extreme LoC 738

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRA parameter ρ (with lower values indicating higher tolerance for risk). *LoC* is the z-score of a single index for locus of control, constructed as detailed in Table A1. Higher values correspond to more external beliefs. All regressions include the relevant LoC terms and adjust for age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), and the Big 5 personality traits. Data: SOEP-IS Sample I3.

Table A10: Non-participation in the survey module

Panel a.) Non-participation in treatment and control				
	Treatment	Control	Diff.	N
Non-participation	0.195 (0.017)	0.165 (0.016)	0.030 (0.023)	1,115
Panel b.) Baseline risk aversion among participating and non-participating respondents				
	Non-participation	Participation	Diff.	N
Baseline Risk	4.925 (0.163)	5.078 (0.078)	-0.152 (0.181)	1,115
Panel c.) Baseline risk aversion of non-participating respondents in treatment and control				
	Treatment	Control	Diff.	N
Baseline Risk	4.927 (0.219)	4.923 (0.244)	0.004 (0.328)	201

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. OLS regressions with robust standard errors in parentheses.

Panel a) reports the likelihood of non-participation in treatment versus control, along with its difference. Panel b) reports the average baseline risk aversion (measured on 10-point of participants and non-participants, along with its difference. Panel c) displays the average baseline risk aversion of non-participating respondents in the treatment and control groups, along with its difference. *Baseline Risk* is measured prior to the treatment on a scale from very willing to take risks (0) to not very willing to take risks (10). Data: SOEP-IS Sample I3.

Table A11: Results for Inverse Probability Re-Weighting

	=1, if Lottery 1		CRRA Parameter	
	(1)	(2)	(3)	(4)
Treated	-0.056** (0.027)	-0.054** (0.027)	-0.602** (0.285)	-0.583** (0.286)
Treated x LoC		-0.097*** (0.028)		-0.937*** (0.296)
LoC		0.037* (0.022)		0.358 (0.238)
Observations	914	914	914	914
Model	LPM	LPM	Interval	Interval

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Linear probability models (Columns 1 and 2) and interval regressions (Columns 3 and 4) with standard errors in parentheses. Observations weighted by the inverse of the probability that they are part of our study sample. The propensity scores stem from a probit regression of an indicator for respondents in the study sample on the following covariates: age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), the Big 5 personality traits, and the treatment indicator. Data: SOEP-IS Sample I3.

Table A12: Heterogeneous Effects – Alternative LoC measures

	CRRRA Parameter	
	(1)	(2)
Treated	-0.565** (0.278)	-0.568** (0.278)
Overall LoC	0.297 (0.2220)	
Treated x Overall LoC	-0.836*** (0.279)	
External Scale		0.259 (0.220)
Treated x External Scale		-0.738*** (0.284)
Internal Scale		0.118 (0.208)
Treated x Internal Scale		-0.306 (0.283)
Observations	914	914

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRRA parameter ρ (with lower values indicating higher tolerance for risk). “External scale,” “Internal scale” and “Overall LoC” are constructed as outlined in Table A1. All indices are standardized. All regressions adjust for age, gender, education, parents’ education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), and the Big 5 personality traits. Data: SOEP-IS Sample I3.

Table A13: Heterogeneous Effects – Other Personality Traits

	CRRRA Parameter						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Neuroticism	Agreeableness	Extraversion	Openness	Conscientiousness	Optimism	Self-Control	
Treated	-0.583** (0.279)	-0.584** (0.279)	-0.585** (0.279)	-0.586** (0.279)	-0.582** (0.279)	-0.580** (0.279)	-0.575** (0.283)
Interaction Effect	-0.136 (0.282)	0.070 (0.279)	0.024 (0.282)	0.286 (0.282)	0.097 (0.278)	-0.267 (0.285)	-0.140 (0.282)
Level Effect	0.318 (0.203)	0.127 (0.202)	-0.014 (0.220)	-0.154 (0.214)	0.029 (0.207)	0.191 (0.206)	0.275 (0.232)
Observations	914	914	914	914	914	914	879

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRRA parameter ρ (with lower values indicating higher tolerance for risk). "Interaction Effect" is the interaction between Treated and the corresponding personality trait indicated on top of a column, while "Level Effect" corresponds to the association between personality trait and dependent variable. All personality traits are standardized to have a mean of zero and a standard deviation of one. Columns 1-5 show the Big 5 traits and columns 6-7 indicate *Optimism* and *Self-Control* measures. *Optimism* indicates how respondents think about their future ("If you think about the future: Are you... (1) optimistic; (2) rather optimistic than pessimistic; (3) rather pessimistic than optimistic; (4) pessimistic?"). We recoded the variable such that higher values reflect more optimism. *Self-Control* is the standardized sum of 13 self-control items, following Tangney et al. (2004). Higher values imply more indicated self-control. Data: SOEP-IS Sample I3.

Table A14: Heterogeneous Effects - Locus of Control & Emotions

	CRRA Parameter				
	(1)	(2)	(3)	(4)	(5)
Treated	-0.564** (0.278)	-0.575** (0.278)	-0.564** (0.277)	-0.561** (0.278)	-0.576** (0.277)
Treated x LoC	-0.914*** (0.280)	-0.906*** (0.279)	-0.911*** (0.279)	-0.908*** (0.280)	-0.892*** (0.278)
LoC	0.335 (0.219)	0.366* (0.220)	0.307 (0.219)	0.347 (0.220)	0.350 (0.219)
Happiness Index	No	Yes	No	No	Yes
Anger	No	No	Yes	No	Yes
Fear	No	No	No	Yes	Yes
Observations	914	914	914	914	914

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRA parameter ρ (with lower values indicating higher tolerance for risk). *LoC* is the z-score of a single index for locus of control, constructed as detailed in Table A1. Higher values correspond to more external beliefs. All regressions adjust for age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), and the Big 5 personality traits. *Emotions* indicate the frequency of feeling *happy*, *sad*, *angry*, and *fearful* in the last four weeks and is measured on a scale from very rarely (1) to very often (5). *Happiness Index* is an unweighted index of happiness and sadness computed as $(happiness - sadness)/2 + 3$ to equal the range of the other emotions (Anger and Fear). Data: SOEP-IS Sample I3.

Table A15: Heterogeneous Effects - Emotions

	CRRRA Parameter		
	(1) Happiness Index	(2) Anger	(3) Fear
Treated	-2.726* (1.396)	-0.337 (0.813)	-0.833 (0.615)
Interaction Effect	0.578 (0.370)	-0.091 (0.279)	0.140 (0.299)
Level Effect	0.119 (0.279)	0.358* (0.201)	-0.229 (0.230)
Observations	914	914	914

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRRA parameter ρ (with lower values indicating higher tolerance for risk). "Interaction Effect" is the interaction between Treated and the corresponding emotion indicated on top of a column, while "Level Effect" corresponds to the association between emotion and the dependent variable. *Emotions* indicate the frequency of feeling *happy*, *sad*, *angry*, and *fearful* in the last four weeks and is measured on a scale from very rarely (1) to very often (5). *Happiness Index* is an unweighted index of happiness and sadness computed as $(happiness - sadness)/2 + 3$ to equal the range of the other emotions (Anger and Fear). Data: SOEP-IS Sample I3.

Table A16: Heterogeneous Effects - Socio-economic Characteristics

	CRRRA Parameter			
	(1) Female	(2) Unemployed	(3) Income	(4) Education
Treated	-0.380 (0.407)	-0.539* (0.286)	-3.531 (4.390)	-1.011 (0.760)
Interaction Effect	-0.383 (0.556)	-0.873 (1.259)	0.389 (0.580)	0.116 (0.192)
Level Effect	0.790** (0.403)	1.164 (0.867)	0.171 (0.934)	-0.096 (0.222)
Observations	914	914	914	914

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Interval regressions with standard errors in parentheses. Dependent variable is the CRRRA parameter ρ (with lower values indicating higher tolerance for risk). "Interaction Effect" is the interaction between Treated and the corresponding socio-demographic characteristic indicated on top of a column, while "Level Effect" corresponds to the association between the socio-demographic characteristic and the dependent variable. *Female* and *Unemployment* are indicators for gender and employment status. *Income* is the log of equivalence scale determined by the square root scale (i.e., we divide the monthly household net income by the square root of the number of household members). *Education* is measured in 7 categories according to the International Standard Classification of Education (ISCED), with higher categories representing a higher level of education. Data: SOEP-IS Sample I3.

Table A17: Main Results When Fully Interacting Treatment with Covariates

	=1, if Lottery 1		CRRA Parameter	
	(1)	(2)	(3)	(4)
Treated x LoC	-0.093*** (0.027)	-0.097*** (0.032)	-0.892*** (0.280)	-0.917*** (0.328)
Observations	914	914	914	914
Treatment Interacted with Covariates	No	Yes	No	Yes

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Columns (1) and (2) present estimates from linear probability models, where the dependent variable is an indicator equal to 1 if the participant chose lottery 1 (payoff of €50 in both states). Columns (3) and (4) present estimates from interval regressions. All regressions adjust for age, gender, education, parents' education, baseline risk attitudes (measured on a scale ranging from 0: very willing to take risks to 10: not at all willing to take risks), equivalized net income, marital status, household size, employment status (self-employed, unemployed, retired), wealth, satisfaction with health, life satisfaction, region (East/West Germany), the optimism proxy ranging from 1 (optimistic) to 4 (pessimistic), the Big 5 personality traits, and the participant's emotional state (see notes of Table A15 for details). Even columns additionally interact the treatment indicator with the full set of covariates. Data: SOEP-IS Sample I3.

Table A18: Inequality and Locus of Control

	Risk Aversion			
	(1)	(2)	(3)	(4)
Inequality: Gini	-1.185** (0.456)	-0.756* (0.431)	-0.169 (0.364)	-0.275 (0.367)
Inequality x LoC	-0.482** (0.200)	-0.322** (0.160)	-0.295*** (0.110)	-0.278** (0.118)
LoC	0.253*** (0.075)	0.161*** (0.060)	0.164*** (0.043)	0.157*** (0.046)
Constant	0.447** (0.171)	-0.264 (0.233)	-0.286* (0.145)	-0.323** (0.132)
Individual Covariates	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes
Survey Year FE	No	No	No	Yes
Observations	145,206	130,068	130,068	130,068
Countries	71	70	70	70
R ²	0.02	0.09	0.11	0.12

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. OLS regressions on the relationship between income inequality and risk aversion using data from the *World Values Survey*. Standard errors, clustered at the country level, in parentheses. The dependent variable is Schwartz's risk-sensation-seeking measure. Higher values imply higher risk aversion. Inequality measured by the national-level Gini coefficient, after tax and transfers. Higher values imply higher inequality (Scale: 0 to 1). Locus of Control (LoC) is proxied by the standardized answers to the question: "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Indicate how much freedom of choice and control you feel you have over the way your life turns out (Reverse coded scale from 1 – "a great deal" – to 10 – "none at all")." Higher values imply more external beliefs. *Individual covariates* include marital status, number children, subjective health status, satisfaction with financial situation of household, gender, age, education, and employment status. *Region FE* include the Arab World/Maghreb states, Latin Europe, post-Soviet countries, Eastern Europe, Nordic/Germanic countries, Latin American, South/East Asia, and Sub-Saharan Africa. The reference category is Anglo-Saxon countries. *Survey Year FE* is an indicator for wave 5 (base is wave 6). Data: World Values Survey: waves 5 and 6.

Table A19: Wealth Categories by Condition – Manipulation Checks

respondi		Bilendi	
Control	Treatment	Control	Treatment
Up to 3,000€	Up to 334,000€	Up to 7,100€	Up to 473,500€
3,001€ to 11,800€	334,001€ to 555,400€	7,101€ to 19,100€	473,501€ to 779,500€
11,801€ to 31,200€	555,401€ to 861,600€	19,101€ to 49,300€	779,501€ to 1,227,900€
31,201€ to 131,000€	861,601€ to 1,292,100€	49,301€ to 196,100€	1,227,900€ to 1,885,000€
More than 131,000€	More than 1,292,100€	More than 196,100€	More than 1,885,000€

Notes: Wealth categories used in the control and treatment condition of the manipulation checks. Upper bounds taken from the 2017 Household Finance and Consumption Survey (HFCS) for respondi (HFCS, Deutsche Bundesbank, 2019) and the 2023 HFCS for Bilendi (HFCS, Deutsche Bundesbank, 2025).

A.6 Survey Module – SOEP-IS

Our survey module consists of three parts: questions on personality traits, the treatment manipulation, and a lottery task to measure risk-taking. In what follows, we list the English wording of each question (translated from German). See here for the publicly available survey instruments: https://www.diw.de/documents/publikationen/73/diw_01.c.789423.de/diw_ssp0866.pdf.

Optimism. We measure optimism with two questions: a general question about optimism regarding the future taken from the SOEP, and a question about the likelihood of experiencing an event relative to an average person:

1. *When you think about the future: are you... [Scale: (i) ...optimistic, (ii) ...rather optimistic than pessimistic, (iii) ...rather pessimistic than optimistic, (iv) ...pessimistic?]*
2. *Compared to other people of the same age and gender as you: how likely is it that throughout your life you will...*
 - *...be financially successful?*
 - *...not suffer from a serious illness?*
 - *...be successful in your job?*
 - *...be satisfied overall?*

For each component, subjects could respond on a scale from 1 to 7, where 1 indicates very much less likely, 4 indicates as likely as the average person, and 7 indicates very much more likely.

Locus of Control. We implemented the same ten items that are routinely used in the SOEP (Nolte et al., 1997). Respondents answered on a Likert scale ranging from 1 (disagree completely) to 7 (agree completely):

The following statements capture different attitudes towards life and the future. To which degree do you personally agree with the statements?

- (i) *The course of my life is depending on me.*
- (ii) *In comparison to others, I have not achieved what I deserve.*

- (iii) *What you achieve in life is first of all a question of fate or luck.*
- (iv) *I often experience that others are deciding about my life.*
- (v) *You have to work hard to be successful.*
- (vi) *When I face difficulties in life, I often doubt my abilities.*
- (vii) *Which opportunities I have in life is determined by social conditions.*
- (viii) *More important than all effort are the abilities you have.*
- (ix) *I have little control over the things happening in my life.*
- (x) *Social or political involvement can influence social conditions.*

Treatment Manipulation. Our treatment manipulation is embedded in a question about respondents' wealth. Specifically, we ask respondents to indicate their net wealth using five predefined wealth categories. To manipulate respondents' perception regarding the distribution of wealth, we randomly vary the available categories. That is, we assign half of the respondents to categories with relatively wide intervals (treatment group), while the other half of the respondents are assigned to much smaller intervals (control group):

Now I would like to talk with you about wealth. One can divide households in Germany into five wealth categories. Wealth in this context refers to net wealth. That is, it is equivalent to total household wealth, including cash, savings accounts, stocks, and real estate, minus debts, such as loans, mortgages, or credit card debt. Please indicate to which category your household belongs:

- (i) *Up to €2,500 (**Treatment Group:** Up to €275,000)*
- (ii) *€2,501 to €11,000 (€275,001 to €468,000)*
- (iii) *€11,001 to €27,000 (€468,001 to €722,000)*
- (iv) *€27,001 to €112,000 (€722,001 to €989,000)*
- (v) *More than €112,001 (More than €989,001)*

Risk Elicitation. In the risk elicitation part, respondents faced two questions. The first question asked them to indicate whether they would like to participate in a lottery game, whereas the second contained the actual lottery choice:

*Now let us turn to a special task with which we try to understand how people make financial decisions. We will present you with six lotteries. We ask you to choose one of the six lotteries. At the end of this block of questions, every 10th respondent will be randomly chosen to actually receive the winnings from their chosen lottery. The actual payment will be made at the end of the complete questionnaire. [Answers: (1) start the lottery game, (2) I do not want to participate because...*open].*

I will now show you six different lotteries on the screen. Each lottery consists of two pay-offs, each of which can be drawn with equal probability. The lotteries differ in the sense that for each lottery you have an equal chance of winning different amounts of money. At the end of this module, we will determine whether your choice will actually be paid out to you. Which lottery do you choose?

- (i) Lottery 1: 50% 50€ / 50% 50€
- (ii) Lottery 2: 50% 45€ / 50% 95€
- (iii) Lottery 3: 50% 40€ / 50% 120€
- (iv) Lottery 4: 50% 30€ / 50% 150€
- (v) Lottery 5: 50% 10€ / 50% 190€
- (vi) Lottery 6: 50% 0€ / 50% 200€

Top-10% income threshold. *In your estimation, what gross annual income do you need to be in the top 10 percent of German households?*